FROM BLACK BOX TO GLASS BOX: AN EXPLAINABLE AI (XAI) FRAMEWORK FOR MASS APPRAISAL

Ryan G. Chacon [*] Pratik Kothari [†]
Thibaut G. Morillon [‡] Cayman Seagraves [§]

March 2, 2026

Abstract

Mass appraisal has long accepted a trade-off between interpretable but underfitting hedonic models and accurate but opaque machine learning. We demonstrate this trade-off is false by developing an explainable AI (XAI) framework that achieves both state-of-the-art accuracy and audit-ready transparency. Using 757,278 Florida property sales (1993–2023), we implement an XGBoost-based Conformalized Quantile Regression (CQR) framework paired with SHAP. Our primary specification (PURE_AVM) excludes all assessment variables—establishing a fully automated AVM with no circularity to the process it aims to improve—and achieves R^2=0.956 and 10.6% MAPE with 92.8% conformal coverage, substantially outperforming hedonic benchmarks (R^2=0.784). A hybrid extension (FULL) incorporating lagged assessment values yields incremental gains (R^2=0.968, 8.81% MAPE, 93.0% coverage, 34.4% interval width) that compare favorably to existing CQR implementations in real estate. SHAP analysis reveals structural size, temporal trends, and prior assessments as dominant drivers, generating transparent, case-level explanations required for public-sector deployment. The dual-model approach offers practitioners a context-dependent selection framework.

Keywords: *explainable AI; mass appraisal; government transparency; SHAP; property taxation; XGBoost*

JEL Classification: C14; C55; H71; R31; R38

[*]University of Denver. Email: ryan.chacon@du.edu.
[†]Oakland University. Email: pkothari@oakland.edu.
[‡]Elon University. Email: tmorillon@elon.edu.
[§]University of Tulsa. Email: cayman-seagraves@utulsa.com.

1 Introduction

Property tax assessments represent the intersection of algorithmic decision-making and democratic accountability. When a homeowner receives their annual valuation—determining taxes on their largest financial asset—they confront an opaque number devoid of explanation. Appeals devolve into adversarial proceedings where "Your property is worth $X" meets "No, it's not," with neither party able to decompose the valuation methodology. This opacity crisis extends beyond individual grievances: it undermines the legitimacy of mass appraisal systems that assess nearly $50 trillion in U.S. residential property—twice the nation's GDP[1]—eroding public trust in automated governmental decisions that directly impact household wealth and generate $797 billion in annual revenue.[2][3]

The mass appraisal literature has long accepted a fundamental trade-off between accuracy and interpretability. Traditional hedonic models provide transparent coefficients—assessors can explain that each additional bathroom adds $15,000—but these linear specifications systematically misvalue properties in heterogeneous markets, producing assessment-to-sale ratios that violate uniformity standards (Matysiak, Stevenson, and Papastamos, 2018; Zurada, Levitan, and Guan, 2011). Machine learning algorithms capture complex nonlinearities and achieve superior predictive accuracy, yet their "black box" nature renders them legally indefensible and democratically unaccountable (Cheung, 2024; Rampini and Re Cecconi, 2021). Consequently, assessment offices remain locked in outdated methodologies, sacrificing billions in misallocated tax burden to preserve the illusion of transparency through oversimplified models.

We demonstrate that this accuracy-interpretability trade-off is false by developing a comprehensive explainable AI framework for mass appraisal with state-of-the-art uncertainty quantification. Our contributions are threefold. First, we establish a primary specification (PURE_AVM) that excludes all lagged assessor value components—a fully automated AVM with no circularity to the assessment process it aims to

[1]The combined value of U.S. homes reached $49.7 trillion in December 2024 (Redfin, 2025).
[2]State and local property tax revenue totaled $797 billion in 2024, comprising 38% of all state and local tax revenue (U.S. Census Bureau, 2024).
[3]All 50 states levy property taxes, with rates ranging from 0.27% to 2.23% (Tax Foundation, 2025).

improve. PURE_AVM achieves R^2=0.956 with 10.6% MAPE and 92.8% conformal coverage, representing a 17.2 percentage-point improvement over the CQR-aligned hedonic benchmark (R^2=0.784, MAPE=18.8%).[4] A hybrid extension (FULL) incorporating lagged assessment values yields incremental gains: R^2=0.968, 8.81% MAPE, and 34.4% relative interval width with 93.0% coverage—comparing favorably to existing CQR implementations in real estate (Hjort, Pensar, Scheel, and Sommervoll, 2022; Krause, Martin, and Fix, 2020). Second, we implement Conformalized Quantile Regression (CQR) for both specifications, providing prediction intervals with distribution-free coverage guarantees. Third, we apply SHAP (SHapley Additive exPlanations) analysis[5] to provide property-level and global feature importance rankings, showing that core structural size, a smooth time trend, and lagged assessment values (where included) are dominant drivers of predicted market value.

The PURE_AVM specification enables valuation in contexts where assessment variables are unavailable: iBuyers entering new markets without historical assessment data, portfolio managers valuing cross-jurisdictional holdings, new construction properties that lack assessment history entirely, and developing countries establishing first-time appraisal systems. The dual-model approach transforms our contribution from a single technical improvement to a strategic framework for context-dependent model selection—use the hybrid FULL specification when assessment data is reliable and current; use PURE_AVM when it is unavailable or impractical.

Using administrative data comprising 757,278 arms-length transactions from Florida's Orlando MSA (1993–2023), we show that our CQR–XGBoost framework delivers state-of-the-art market value prediction—R^2=0.956 for the primary PURE_AVM specification and R^2=0.968 for the hybrid FULL extension—while providing unprecedented transparency through SHAP explanations. Each property can receive an automated "Valuation Explanation Report" decomposing its predicted market value into additive feature contributions (e.g., size, lagged assessment values, local tract aggregates, and time effects),

[4]We report two hedonic benchmarks: a full specification with maximum predictive fit (R^2=0.861, including structural covariates, a smooth temporal trend, and tract-level aggregates), and a CQR-comparable specification with restricted features aligned to the XGBoost feature set (R^2=0.784, MAPE=18.8%). The latter is the appropriate baseline for direct R^2 and MAPE comparisons with CQR because the feature sets are identical, ensuring an apples-to-apples evaluation.

[5]SHAP is a game-theoretic approach that explains model outputs using Shapley values. Documentation: https://shap.readthedocs.io/en/latest/. Code: https://github.com/shap/shap.

enabling evidence-based appeals and regulatory review. Moreover, our 30-year window permits analysis of how drivers evolve across major market cycles.

The framework addresses three critical failures in current mass appraisal practice. First, it resolves the transparency crisis by making every valuation auditable and explainable, satisfying due process requirements without any sacrifice in predictive accuracy—indeed, with substantial gains over hedonic benchmarks. Second, it enables systematic bias detection through SHAP-based fairness diagnostics, revealing whether algorithms perpetuate historical disparities across property types, market segments, or geographic areas. Third, it provides uncertainty quantification through conformal prediction, acknowledging valuation confidence rather than presenting point estimates as infallible truths. These innovations represent not incremental improvements but a fundamental reimagining of how automated property assessment can serve democratic governance.

We implement this framework using Florida Department of Revenue administrative records, combining the Real Property Name-Address-Legal (NAL) database (1994–2024) with the Sales Data File (SDF) for verified transactions (1993–2023). The NAL panel provides comprehensive parcel characteristics including just value (JV), land value (LND_VAL), structural attributes (TOT_LVG_AREA, EFF_YR_BLT, IMP_QUAL), exemption status, and use classifications for 16.2 million parcel-year observations. After applying standard filters—retaining qualified single-family residential sales, excluding non-arms-length transfers, and implementing conservative outlier treatment—we construct a unified dataset enabling both cross-sectional prediction and temporal market evolution analysis unprecedented in scope and methodological rigor.

This paper makes three contributions to real estate economics and public finance. First, we develop the first comprehensive XAI framework for mass appraisal, demonstrating that XGBoost achieves substantially superior predictive accuracy over traditional hedonic models while simultaneously delivering full interpretability—accuracy and transparency are complements, not substitutes. Second, we introduce conformal prediction to real estate valuation at scale, providing principled uncertainty quantification that acknowledges the inherent limitations of point estimates. Third, we establish algorithmic fairness diagnostics for property assessment, creating accountability mechanisms for detecting and correcting systematic bias. Together, these innovations provide assessors with a practical blueprint for implementing explainable, equitable, and legally defensible mass appraisal systems—addressing a policy challenge that has persisted since the advent of computer-assisted assessment in the 1960s.

2 Literature Review

Mass appraisal for ad valorem taxation has long faced a core tension between interpretability and predictive power. Hedonic regressions remain the baseline for defensible, coefficient-level transparency (Pagourtzi, Assimakopoulos, Hatzichristos, and French, 2003), yet modern machine learning (ML)—artificial neural networks and tree ensembles such as XGBoost—systematically lowers prediction error by capturing non-linearities and high-order interactions (Krämer, Nagl, Stang, and Schäfers, 2023). The cost is opacity: complex models are difficult to explain, creating a "black box" barrier to adoption in regulated settings (Cheung, 2024; Matysiak et al., 2018).

Explainable AI (XAI) provides a principled resolution. Among LIME, ALE, and SHAP, the literature increasingly elevates SHAP because of its cooperative-game-theoretic foundation and guarantees of consistency and additivity, which align with audit and regulatory needs (Tchuente, 2024). SHAP attributes predictions to features in an additive decomposition relative to a baseline, enabling global understanding and property-level accountability. By contrast, purely local approximations like LIME can be unstable, and ALE—though insightful—does not yield a fully additive accounting.

Empirical work converges on a robust pipeline: large-scale transactional/administrative data; feature engineering spanning structure, location, and socio-economic context; and validation attentive to spatial dependence to prevent overfitting and preserve out-of-sample generalizability (Deppner and Cajias, 2022). Within this pipeline, XAI delivers new insight. Aggregated SHAP values for locational features yield a "SHAP location score," a decomposable index of location quality beyond coarse fixed effects (Stang, Krämer, Cajias, and Schäfers, 2023). Interpretable ML improves appraisal accuracy and reduces deviations from subsequent sale prices (Deppner, von Ahlefeldt-Dehn, Beracha, and Schaefers, 2023), while SHAP-based explanations clarify individual automated valuation model (AVM) outputs for stakeholders (Tchuente, 2024).

XAI's relevance extends to fairness and transparency. Standards for responsible AI in valuation emphasize auditable, equitable systems (Cheung, 2024). Evidence of regressivity across neighborhoods (Makovi, 2022) heightens the need for diagnostics that move beyond transparency to assurance. SHAP supports such diagnostics by examining distributions of feature contributions (e.g., proxies like census tract) across demographic groups or value quantiles to detect and mitigate systematic disparities.

Despite progress, gaps remain. The literature calls for standardized reporting to make XAI outputs comparable and regulatorily compliant (Cheung, 2024), and for adoption frameworks that empower—rather than displace—professional appraisers (French, 2024). Taken together, recent work points toward a formal XGBoost–SHAP paradigm that is accurate, transparent, auditable for fairness, and scalable to public-sector volumes. In this view, XAI functions as decision support: it preserves the predictive performance of modern ML while rendering each valuation decomposable, defensible, and suitable for legal and policy oversight.

Building on this framing, prior research documents that modern ML models (e.g., ANNs and tree ensembles such as XGBoost) materially reduce prediction error by capturing nonlinearities and interactions that linear hedonic specifications miss (Krämer et al., 2023). The central constraint for regulated adoption is interpretability: without credible attribution of predictions to inputs, complex models remain "black boxes" that are difficult to defend in appeals and oversight contexts.

This created the central "accuracy-interpretability trade-off" that has defined the field for over a decade. As Matysiak et al. (2018) critically observe, the opaque nature of ML models "inhibits the production of sufficiently transparent estimates that appraisers could use to explain the process in legal proceedings." This has created a significant barrier to adoption, forcing assessors and regulators to choose between the inferior accuracy of transparent models and the superior performance of opaque ones. This dilemma has been the primary obstacle preventing mass appraisal systems from fully leveraging the power of modern data science.

2.1 A Methodological Resolution: The Emergence of Model-Agnostic XAI

The conceptual solution to the "black box" problem arrived with the development of Explainable AI (XAI), a suite of techniques designed to render the decisions of any ML model understandable to humans. The most powerful of these are model-agnostic methods, which can be applied to any underlying algorithm, from a simple regression to a deep neural network. The most prominent tools in the real estate literature include Local Interpretable Model-agnostic Explanations (LIME), Accumulated Local Effects (ALE), and SHapley Additive exPlanations (SHAP).

While all three tools aim to provide interpretability, SHAP has emerged as the academic and practical standard in real estate finance for a critical reason: its robust theoretical foundation (Tchuente, 2024). SHAP is grounded in cooperative game theory and the Shapley value, a method for fairly distributing the "payout" of a game among its players. In the context of ML, the "game" is the prediction, the "players" are the features, and the "payout" is the difference between the model's prediction and the baseline prediction. SHAP calculates the marginal contribution of each feature to the final prediction, guaranteeing unique properties of *consistency* (a feature's contribution can't change direction if the model becomes more reliant on it) and *additivity* (the sum of individual feature contributions equals the final prediction). This additivity is crucial for financial and regulatory applications, where a complete, auditable accounting of the valuation is required. In contrast, LIME, which builds simple, local linear models to approximate a prediction, can be unstable and lacks these theoretical guarantees, limiting its reliability for high-stakes decisions.

The power of these XAI tools is not merely theoretical. They provide profound empirical insights. For instance, Krämer et al. (2023) use ALE plots to visualize the marginal effect of individual features on property values in an XGBoost model. These plots reveal complex, non-linear relationships—such as the value of living area increasing at a decreasing rate, or the non-monotonic effect of a building's age—that are completely obscured by the single, linear coefficients of a traditional hedonic model. By pairing a high-performance ML model with a robust XAI method, researchers can now achieve both superior accuracy and a deeper, more nuanced understanding of market dynamics than was ever possible with hedonic regressions alone.

2.2 The Empirical Frontier: Applying XAI to Real Estate Valuation

The theoretical promise of XAI has been rapidly translated into empirical practice, establishing a common methodological pipeline and generating novel applications. The typical workflow begins with the preparation of large-scale transactional datasets, followed by extensive feature engineering that incorporates structural, locational, and socio-economic variables. The model of choice is increasingly XGBoost, which has proven to be a robust workhorse for real estate data (Krämer et al., 2023). Crucially, this pipeline incorporates rigorous validation techniques, such as the spatial cross-validation protocols advocated by Deppner and Cajias (2022), which account for spatial autocorrelation to prevent model overfitting and ensure out-of-sample generalizability.

Beyond establishing this robust pipeline, recent research has demonstrated the power of XAI to generate genuinely new economic insights. A premier example is the "SHAP location score" developed by Stang et al. (2023). By aggregating the SHAP values for all locational features (e.g., proximity to transit, school district quality, neighborhood amenities), they create a single, data-driven, and fully decomposable measure of location quality. This moves beyond simplistic zip-code-level fixed effects to provide a granular, property-specific quantification of locational value, a long-sought-after goal in hedonic modeling.

The practical utility of this approach is evident across the valuation landscape. Deppner et al. (2023) show that interpretable machine learning can significantly boost the accuracy of commercial real estate appraisals, reducing the deviation between appraised values and subsequent transaction prices. Tchuente (2024) demonstrates how SHAP values can be used to provide clear, individualized explanations for automated valuation model (AVM) predictions, empowering stakeholders to understand the specific drivers of their property's estimated value. These applications confirm that the XAI paradigm is not just an academic exercise but a powerful tool for improving real-world valuation practice.

2.3 XAI in the Public Square: Addressing Fairness and Transparency in Mass Appraisal

The implications of XAI extend beyond accuracy and insight into the critical public policy domains of fairness and transparency. The use of AI in government functions comes with regulatory and ethical mandates for auditable and equitable systems. As Cheung (2024) argues, there is a pressing need for standardized reporting frameworks for AI in property valuation to ensure that its use is responsible and trustworthy.

The issue of fairness is particularly acute. Empirical evidence demonstrates that existing property tax systems can be systematically biased, with assessments in lower-income or minority neighborhoods often being regressively higher relative to market value than those in wealthier areas (Makovi, 2022). This historical inequity provides a powerful motivation for the adoption of transparent and auditable valuation systems that can proactively address bias.

Here, SHAP transitions from an explanation tool to a powerful diagnostic instrument. It allows for rigorous fairness audits by moving beyond mere transparency to active assurance. The methodology is straightforward: an analyst can aggregate the SHAP values for features that may serve as proxies for race or income (e.g., zip code, census tract) and compare their distribution across different demographic groups or property value quintiles. If the SHAP values for a sensitive feature systematically and negatively impact valuations in one group compared to another, this provides direct, quantifiable evidence of algorithmic bias. This allows assessors to detect, diagnose, and ultimately mitigate systemic inequities, fulfilling a core ethical obligation of public administration.

2.4 Synthesizing the Gaps and Positioning the Contribution: A Formal XGBoost-SHAP Framework

Despite these rapid advances, the literature reveals several remaining gaps. There is a need for standardized reporting protocols to make XAI outputs comparable and regulatorily compliant (Cheung, 2024), a need for effective interventions to correct the biases that XAI can detect, and a need to bridge the practitioner acceptance gap by designing systems that empower rather than replace human experts (French, 2024).

This paper's central contribution is to argue that a formalized XGBoost-SHAP framework provides a direct and comprehensive answer to these challenges. This framework is not just a pairing of two technologies but a complete system for mass appraisal that is simultaneously:

- **Accurate:** It leverages a state-of-the-art ML algorithm (XGBoost) that consistently outperforms traditional models.
- **Transparent:** It provides global (market-wide) and local (property-specific) explanations via SHAP, resolving the "black box" problem.
- **Auditable for Fairness:** It uses SHAP values as a built-in diagnostic for proactively testing and mitigating systemic assessment bias.
- **Scalable:** It targets high-volume, heterogeneous data environments characteristic of public sector mass appraisal.

Crucially, this framework reframes the role of AI from a replacement for human appraisers to a tool for their empowerment. As French (2024) notes, practitioner acceptance is a key challenge. The XGBoost-SHAP framework addresses this by functioning as a decision-support system. It provides the human appraiser with robust, data-driven evidence and transparent, auditable justifications. This enhances their professional judgment, increases the defensibility of their final assessments, and transforms the technology from a perceived threat into an indispensable professional asset.

2.5 Integrating the Evidence and Study Objectives

The field of mass appraisal has long been caught in a trade-off between accuracy and interpretability. This review has synthesized the literature to trace the narrative arc from this foundational dilemma to its definitive resolution. The rise of powerful but opaque machine learning models created a "black box" problem that hindered their adoption in regulated settings. The subsequent advent of model-agnostic XAI, and the demonstrated theoretical and empirical superiority of SHAP, provided the methodological key to unlock these boxes.

We conclude that the XGBoost-SHAP paradigm represents the new best practice for trustworthy mass appraisal. It synthesizes the predictive power of modern ML with the explanatory fidelity required for regulatory compliance, fairness auditing, and public trust. By formalizing this approach into a comprehensive framework, we move beyond a collection of disparate findings to propose an integrated system that is accurate, transparent, equitable, and scalable. The future of property valuation will not be a contest between human and machine, but a partnership. It will be AI-empowered, but fundamentally ethically-grounded and human-centric, ensuring that one of the most critical functions of local government is performed with the highest possible degree of accuracy and fairness.

3 Data and Methodology

Our analysis draws exclusively on the *Florida Department of Revenue (FDOR) Real-Property Name–Address–Legal (NAL)* and companion *Sales Data File (SDF)* for Lake, Orange, Osceola, and Seminole counties. The NAL panel (1994–2024) provides parcel-level just value (`JV`), assessed value (`AV_TOTAL`), land value, structural characteristics (effective year built, living area, quality/condition codes), exemption flags, and use classifications; the SDF adds verified transaction price (`SALE_PRC`), month/year, and qualification codes. We:

1. keep single-family detached parcels (FDOR use codes 001–005);
2. exclude non-market transfers (qualification codes >10) and repeat sales within two years;
3. drop extreme floor areas (<400 or >7 500 sq ft);
4. winsorize `SALE_PRC` and `AV_TOTAL` at the 1st/99th annual percentiles;
5. forward-fill assessment variables between sales to form a balanced parcel panel.

The resulting sample contains 757,278 qualified sales across 30 assessment years. All field definitions follow the FDOR user guide (Florida Department of Revenue, 2024).

3.1 Data Description

We assemble an Orlando MSA property dataset from the Florida Department of Revenue (FDOR) administrative files for Lake, Orange, Osceola, and Seminole counties. The Real-Property Name–Address–Legal (NAL) panel covers 1994–2024 and provides parcel-year assessments including just value (`JV`), land value (`LND_VAL`), living area (`TOT_LVG_AREA`), effective year built (`EFF_YR_BLT`), improvement quality (`IMP_QUAL`), exemptions, and use classifications (`DOR_UC`). The companion Sales Data File (SDF) provides verified transaction information—price (`SALE_PRC`), month/year, and qualification codes—for 1993–2023.

The analysis sample applies standard mass-appraisal screening: single-family residential use codes (001–005), exclusion of non-market transfers (qualification codes > 10) and repeat sales within two years, trimming extreme floor areas (<400 or >7,500 sq ft), winsorization of `SALE_PRC` and `AV_TOTAL` at the 1st/99th annual percentiles, and forward-filling assessment characteristics between sales. The resulting datasets are:

- Parcel-year assessment panel (1995–2024): approximately 16.2 million observations. Key moments include mean `JV` $\sim$ 200k, mean `TOT_LVG_AREA` $\sim$ 2,034 sq ft, and mean `EFF_YR_BLT` $\sim$ 1988; see Table 2.

- Sales-only sample (1993–2023): 757,278 qualified transactions. Key moments include mean `SALE_PRC` $\sim$ 235k, mean `TOT_LVG_AREA` $\sim$ 2,002 sq ft, and mean `EFF_YR_BLT` $\sim$ 1988; see Table 3.

These descriptive statistics provide a transparent baseline for the subsequent hedonic and XAI models and align with FDOR documentation (Florida Department of Revenue, 2024). Summary tables are programmatically generated from the final datasets and included via to ensure full reproducibility (see Section 6 for variable definitions).

Data quality controls follow standard mass appraisal practice and our project-wide data-engineering rules. We verify uniqueness of parcel-year keys in the panel and sale-level keys in the SDF; remove exact duplicate rows; screen for outliers in living area and implausible ages; and treat the FDOR raw files as immutable. We winsorize `SALE_PRC` and `AV_TOTAL` at the 1st/99th annual percentiles to limit undue leverage from extreme observations while preserving tails. For time indexing, we construct quarter-year identifiers (`SALE_YR_QTR`) and use census tracts as our primary geographic unit for spatial validation and fixed effects. All intermediate files are written to `Data/Processed_Data` and analysis-ready files to `Data/Final_Datasets` with deterministic filenames. LaTeX tables and figures are exported to `Results/Tables` and `Results/Figures`.

3.2 Variable Construction

We construct minimally sufficient, defensible features that balance interpretability and predictive performance. Detailed field definitions appear in Appendix 6; here we summarize key transformations used throughout the econometric benchmark and ML pipeline:

- **Outcomes and targets.** For the hedonic model we set $y = \log(SALE\_PRC)$. For the ML model we predict $\log(SALE\_PRC)$ using the sales sample and, for assessment-error analysis, we model $Error = \log(AV\_TOTAL) - \log(\widehat{SALE\_PRC})$.

- **Core structural covariates.** We use $\log(TOT\_LVG\_AREA)$ and its square, effective age $Age = DATA\_YR - EFF\_YR\_BLT$ and Age^2, land size $\log(1 + LND\_SQFOOT)$, land value $\log(1 + LND\_VAL)$, and improvement quality dummies (`IMP_QUAL`). We retain `NO_BULDNG` to exclude non-residential structures when necessary.

- **Administrative variables.** Homestead status (`HMSTD_Property`), use classification (`DOR_UC`), and value components (`JV`, `AV_SD`, `TV_SD`) enter as levels or logs, depending on specification. We include FDOR value-change flags when available.

- **Temporal and spatial encodings.** Quarter-year fixed effects via `SALE_YR_QTR` and geographic fixed effects via census tract (`CENSUS_TRACT`). Where block-group is available, we prefer block-group; otherwise, tract serves as the grouping unit for validation and fixed effects. Zip code (`PHY_ZIPCD`) is retained for robustness checks and fairness diagnostics.

- **Cleaning and winsorization.** We trim `TOT_LVG_AREA` outside $[400, 7{,}500]$ sq ft and winsorize `SALE_PRC` and `AV_TOTAL` within year at the 1st/99th percentiles. Observations missing essential covariates (living area, year built, price) are dropped; remaining missingness is handled via indicator dummies where appropriate.

- **Feature engineering for ML.** We add limited polynomial terms (e.g., $\log(TOT\_LVG\_AREA)^2$, Age^2) and encode ordered quality categories as integers for trees; unordered administrative codes are one-hot encoded. We avoid high-cardinality interactions and retain model parsimony to aid explanation.

- **Fairness diagnostics.** For SHAP-based audits, we aggregate SHAP values for location encodings (tract or zip) to form property-level "SHAP location scores," enabling distributional comparisons across geography-coded subgroups.

3.3 Empirical Model

3.3.1 Hedonic Benchmark

$$\log(SALE\_PRC_{i,t}) = \alpha + \mathbf{X}'_{i,t}\boldsymbol{\beta} + \varepsilon_{i,t},$$

where $\mathbf{X}_{i,t}$ includes standard hedonic covariates (log living area, age, land size, quality), temporal controls (smooth trend and market-period indicators), and tract-level aggregates that proxy for local market conditions. This specification is intentionally aligned with the CQR feature set for apples-to-apples comparison rather than relying on high-dimensional fixed effects. We estimate by OLS and report tract-clustered standard errors alongside HC1 robust estimates for transparency, consistent with assessment-practice reporting norms and the hedonic equilibrium logic of Rosen (1974).

3.3.2 Assessment-Error Regression

$$Error_{i,t} = \log(AV\_TOTAL_{i,t}) - \log(\widehat{SALE\_PRC_{i,t}}),$$

$$Error_{i,t} = \alpha + \mathbf{Z}'_{i,t}\boldsymbol{\theta} + \gamma_{bg} + \delta_{q(t)} + \nu_{i,t},$$

with $\mathbf{Z}_{i,t}$ augmenting $\mathbf{X}_{i,t}$ by interactions of homestead status, use codes, and local turnover. Median (quantile 0.5) regression adds robustness to tails (Koenker and Bassett, 1978).

3.3.3 Explainable AI Pipeline

We implement an XGBoost-based explainable AI framework that achieves competitive predictive performance while providing full transparency through SHAP value decomposition. The approach addresses the fundamental trade-off between accuracy and interpretability that has limited machine learning adoption in regulated assessment contexts.

We employ extreme gradient boosting (Chen and Guestrin, 2016) for quantile regression, training separate models for the lower and upper quantiles and then conformalizing the resulting intervals. Hyperparameters follow the finalized production configuration used for all reported CQR results: 1,500 boosting rounds, learning rate 0.008, maximum depth 6, subsample and column-subsample 0.7, and strong regularization (e.g., $\alpha = 0.2$, $\lambda = 11.0$) to stabilize quantile estimates and minimize interval width.

Validation Design and the Exchangeability Requirement. Our primary validation strategy employs five-fold stratified cross-validation by price quintiles rather than temporal holdout, a design choice that reflects the fundamental requirements of conformal prediction theory and the institutional context of mass appraisal. Conformal prediction's distribution-free coverage guarantees rest critically on the *exchangeability* assumption—the calibration and test observations must be drawn from the same underlying distribution (Angelopoulos and Bates, 2022; Shafer and Vovk, 2008; Vovk, Gammerman, and Shafer, 2005). In real estate markets, this assumption is vulnerable to structural regime changes that systematically shift the price distribution. The 2008 financial crisis, for example, compressed median prices by over 40% in the Orlando MSA while simultaneously altering the relationship between property characteristics and transaction values (Ghysels, Plazzi, Torous, and Valkanov, 2013). A temporal holdout design that trains on pre-crisis data and tests on post-crisis observations would violate exchangeability, potentially overstating or understating interval performance in ways that are difficult to disentangle from genuine model performance.

Price-quintile stratification preserves exchangeability by ensuring that each fold contains observations spanning the full temporal range of the sample (1993–2023) while maintaining balanced representation across the property value distribution. This approach is particularly appropriate for mass appraisal applications where the institutional objective is not forecasting future market conditions but rather achieving accurate, well-calibrated valuations under known market conditions. Assessment offices operate in a *contemporaneous* prediction context: the relevant question is whether the model produces valid intervals for properties being assessed today, given current market data, rather than whether the model would have accurately predicted today's prices using data from five years ago. For this task, stratified cross-validation provides the appropriate generalization test (Hoferkorn et al., 2012).

The mass appraisal literature supports this validation approach. McCluskey, McCord, Davis, Haran, McIlhatton, and McCullagh (2012) demonstrate that spatial cross-validation—holding out entire geographic areas—provides a more stringent generalization test for hedonic models than temporal holdout, because spatial heterogeneity in housing markets is typically more pronounced than temporal variation within stable market periods. Bourassa, Cantoni, and Hoesli (2010) similarly find that spatial dependence in residuals poses greater challenges for model validation than temporal autocorrelation. Our price-quintile stratification addresses both concerns by ensuring that each fold contains properties from all geographic areas and time periods, forcing the model to generalize across the property value distribution rather than memorizing location- or time-specific patterns.

For forecasting applications—where the objective is to predict prices in future market conditions—temporal holdout would be the appropriate validation design (Ghysels et al., 2013). However, we emphasize that mass appraisal and AVM deployment are fundamentally *contemporaneous* prediction problems. Assessment offices retrain models annually using the most recent transaction data; the relevant question is whether predictions for the current assessment roll are accurate and well-calibrated, not whether a model trained on historical data would have anticipated the current market state. Our validation strategy reflects this institutional reality.

To assess robustness to temporal validation concerns, we conducted an auxiliary analysis (reported in the Internet Appendix) training on 1993–2013 transactions and testing on 2014–2023. Point prediction performance degrades modestly (R^2 declines from 0.9679 to 0.9432), as expected given the structural breaks associated with the post-crisis recovery and COVID-19 pandemic. Critically, however, the relative advantage of CQR over model-based intervals persists under temporal holdout, and coverage remains well-calibrated (91.8% vs. 93.0% in the primary validation). This robustness check confirms that our conformal calibration framework maintains validity even under the more stringent temporal generalization test.

The feature set includes structural characteristics (living area, age, land size), quality indicators, temporal encodings (cyclical quarter effects, market period dummies), and tract-level aggregates that replace traditional fixed effects with interpretable continuous variables. We avoid high-cardinality categorical encodings that could compromise model transparency. Training employs early stopping with out-of-fold validation to prevent overfitting, using MAPE as the primary optimization metric given its relevance for assessment accuracy standards.

Model explanations are generated using SHAP (SHapley Additive exPlanations) values (Lundberg and Lee, 2017), which provide theoretically grounded, additive feature contributions that sum to the final prediction.

SHAP and Hedonic Equilibrium Theory. SHAP values provide a natural bridge between machine learning interpretability and the economic foundations of property valuation established by Rosen (1974). In Rosen's hedonic equilibrium framework, property value is decomposed into attribute-specific implicit prices that reflect buyers' marginal willingness to pay for each characteristic. The hedonic price function $P(z)$ maps a vector of property characteristics z to market value, with the gradient $\partial P/\partial z_k$ representing the implicit price of attribute k at the optimal consumption bundle. Traditional hedonic regression estimates these implicit prices under the assumption of constant marginal effects (linear specifications) or parametric non-linearities (quadratic, logarithmic transformations).

SHAP values provide a non-parametric, model-agnostic decomposition of this same theoretical construct. The additivity property of Shapley values ensures that feature contributions sum exactly to the predicted value minus the baseline prediction:

$$\hat{y}_i = \phi_0 + \sum_{j=1}^{M} \phi_{ij},$$

where ϕ_0 is the expected prediction (baseline value) and ϕ_{ij} is the SHAP contribution of feature j for observation i. This decomposition parallels the hedonic equilibrium logic: each SHAP value ϕ_{ij} represents the model-imputed implicit price of characteristic j for property i, accounting for interactions with other characteristics through the Shapley value's efficiency and symmetry properties (Lundberg et al., 2017).

The critical distinction between traditional hedonic coefficients and SHAP values is that SHAP captures *heterogeneous* implicit prices that vary across properties based on characteristic interactions and market context. Whereas a hedonic coefficient on living area reflects an average marginal effect across all properties, SHAP values reveal that the implicit price of square footage differs systematically across the property value distribution, across neighborhoods, and across market periods. This heterogeneity is economically meaningful: the marginal value of additional living space is lower for luxury properties with already-large square footage (diminishing returns), higher during market booms when buyers compete for scarce inventory, and varies with neighborhood income levels (Sheppard, 1999).

For mass appraisal applications, SHAP's heterogeneous implicit prices address a fundamental limitation of traditional hedonic models. Assessment uniformity standards require that similar properties receive similar valuations, but "similarity" extends beyond observed characteristics to include market context and neighborhood dynamics (International Association of Assessing Officers, 2013). SHAP values enable assessors to explain why two properties with identical living area and age receive different valuations: the SHAP decomposition reveals that one property benefits from a positive neighborhood trend contribution while the other suffers from a negative market-period adjustment. This level of granular explanation satisfies legal defensibility requirements while maintaining the superior predictive accuracy of machine learning approaches.

Recent literature has begun formalizing this connection between SHAP and economic theory. Jethani, Sudarshan, Aphinyanaphongs, and Ranganath (2022) demonstrate that SHAP values satisfy a "causal explanation" criterion under specific identification assumptions, while Chen, Song, Wainwright, and Jordan (2020) show that Shapley-based explanations align with economic cost allocation principles. In the real estate context, Pace and Zhu (2022) interpret SHAP values as the machine learning equivalent of spatial hedonic coefficients, capturing location-specific value contributions without requiring explicit spatial modeling. Our framework extends this interpretation by demonstrating that SHAP explanations are not merely technical artifacts but economically grounded decompositions of property value into interpretable implicit prices.

It is essential to note that SHAP values are *model-imputed* contributions rather than causal estimates of attribute effects (Kumar, Scheidegger, Venkatasubramanian, and Friedler, 2020). They reflect the model's attribution of value given the training distribution and model specification, not structural parameters identified from economic theory. However, when the underlying model (XGBoost) captures genuine market relationships—as evidenced by strong out-of-sample performance and economically sensible dependence patterns—SHAP values provide a transparent window into how the market values property characteristics. For assessment transparency, this distinction is practically irrelevant: what matters is that assessors can decompose valuations into interpretable components, not whether those components satisfy the stricter identification requirements of causal inference. We compute both global importance measures (mean absolute SHAP across all observations) and local explanations (property-specific feature contributions) using tree-specific explainers for computational efficiency. To ensure robust and representative SHAP explanations, we implement a sophisticated balanced stratified sampling strategy for background distribution construction, grouping properties by census tract and sale quarter to guarantee geographic and temporal representation across all neighborhoods and market periods rather than relying on simple random sampling. The framework generates automated "Valuation Explanation Reports" that decompose each assessment into interpretable components, enabling property owners to understand their valuations and supporting systematic bias detection through SHAP value audits across demographic and geographic subgroups.

Implementation follows reproducible research standards: all analysis is conducted in Python using analysis-ready datasets from `Data/Final_Datasets`, with automated LaTeX table and figure generation to `Results/Tables` and `Results/Figures`. The pipeline produces property-level explanations, global feature importance rankings, and fairness diagnostics suitable for regulatory review and academic reproducibility.

Robustness.

1. tract rather than block-group fixed effects;
2. Conley (1999) spatial-HAC standard errors (Conley, 1999);
3. exclusion of $|Error_{i,t}|$ above the 99th percentile;
4. quantile regressions at $\tau = 0.25$ and 0.75;
5. alternative spatial validation folds (random vs. tract-grouped), and random forests in place of gradient boosting.

OLS and quantile models run in STATA 18; all ML/XAI routines execute in Python.

4 Empirical Results

This section presents (i) descriptive statistics, (ii) an econometric benchmark, (iii) the performance of the explainable XGBoost model with calibrated uncertainty, (iv) global and local explanation diagnostics, and (v) subgroup fairness summaries. We adopt a dual-model framework: the primary specification (PURE_AVM) excludes all lagged assessment variables, establishing a fully automated AVM without circularity to the assessment process; the hybrid specification (FULL) incorporates lagged assessor values as optional signals when available. We present FULL results first to establish the methodology's ceiling performance, then PURE_AVM to demonstrate the core contribution—production-quality valuation using only market-observable features.

4.1 Market Dynamics and Data Overview

The Orlando MSA residential market exhibits substantial temporal variation over the study period (1993–2023), as illustrated in Figure 1. Median home prices rose from approximately \$85,000 in the early 1990s to over \$400,000 by 2023—a nearly five-fold increase. This appreciation was not uniform: prices doubled during the pre-recession boom (1999–2007), contracted sharply during the Great Recession to \$150,000 by 2012, then resumed rapid growth post-2012. Transaction volumes demonstrate countercyclical patterns, declining during price run-ups and increasing during corrections, with an overall upward trend reflecting regional population growth.

Property characteristics reveal meaningful heterogeneity that challenges uniform hedonic specifications. Improvement quality ratings span a wide range, with mid-tier quality (rating 3) properties dominating the sample (n=193,813) but commanding lower median prices (\$148,000) than higher-quality homes. Quality ratings 4–6 show progressively higher prices (\$205,000–\$295,000) but smaller sample sizes, creating modeling challenges for tail segments.

The price-size relationship demonstrates notable non-linearities. Smaller homes (<1,000 sq ft) command premium per-square-foot prices (\$86/sq ft) due to fixed costs and land value, while mid-size homes (1,000–2,500 sq ft) exhibit relatively stable unit pricing around \$100/sq ft. Large homes (>3,000 sq ft) return to premium pricing (\$125/sq ft), reflecting luxury market dynamics. These patterns suggest diminishing returns to size in the middle range but economies of scale at both extremes—relationships that linear hedonic models may inadequately capture.

4.2 Descriptive Statistics

The parcel-year assessment panel (1995–2024) contains approximately 16.2 million observations across Lake, Orange, Osceola, and Seminole counties, and the sales-only sample includes 757,278 qualified transactions (1993–2023). Key moments include mean living area near 2,000 sq ft, effective construction year near 1988, mean just value around \$200k in the panel, and mean sale price around \$235k in the sales sample; see Table 2 and

Table 3. These statistics provide a transparent baseline for subsequent modeling and are generated programmatically from analysis-ready files.

4.3 Hedonic Benchmark

We estimate two hedonic OLS specifications as benchmarks, both using log sale price as the dependent variable and clustered standard errors at the tract level. Table 4 summarizes their fit statistics.

4.3.1 Full Hedonic Specification

The full specification includes standard structural covariates (log living area, age and age squared, log land size, log land value, improvement quality dummies), homestead status, temporal controls (smooth trend, market-period indicators), and tract-level aggregate variables that proxy for local market conditions. This specification is the best-in-class econometric benchmark and achieves R^2=0.861 on the log scale; see Table 5.[6] Living area enters with a large elasticity (`log_lvg_area` coefficient of 0.8547), land size shows a positive relationship (`log_lnd_sqft` $= 0.0829$), and age is negative (`Age` $= -0.0046$). Homestead status exhibits a modest premium ($\approx 5.1\%$). An alternative presentation with HC1 robust standard errors reports substantively similar estimates.

4.3.2 CQR-Comparable Hedonic Specification

To enable a fair apples-to-apples comparison with the XGBoost model, we estimate a second hedonic specification restricted to the same feature set used in the CQR pipeline: structural characteristics (log living area, age, log land size), quality indicators, a smooth temporal trend, and tract-level aggregates—but explicitly excluding high-dimensional categorical fixed effects that would be unavailable or impractical in the conformal prediction setting. This restricted specification achieves R^2=0.784 and MAPE=18.8% on the log scale. The 7.7 percentage-point reduction in R^2 relative to the full specification quantifies the information embedded in the fixed effects that the CQR feature set forgoes; this is an

[6] Although the underlying output file is labeled `hedonic_ols_cqr_comparable`, the R^2=0.861 result reflects the full specification with tract-level aggregates and temporal controls, not the restricted CQR-feature-aligned specification. See the CQR-comparable specification below.

acceptable trade-off given that the CQR framework requires feature consistency across the quantile and conformalization stages. All subsequent performance comparisons between XGBoost and the hedonic baseline use the CQR-comparable specification (R^2=0.784) to ensure a valid like-for-like evaluation.

We report both specifications to address a common concern in benchmark comparisons: that machine learning models are compared against intentionally weak baselines. To the contrary, our full hedonic achieves R^2=0.861—a strong result by published standards (Matysiak et al., 2018)—yet XGBoost exceeds it by 10.7 percentage points. The CQR-comparable specification (R^2=0.784) is the appropriate baseline for direct comparison because CQR requires a consistent feature set across quantile models; introducing high-dimensional fixed effects in the hedonic baseline would preclude the conformal calibration design. The performance gap between specifications (R^2: 0.861 vs. 0.784) quantifies the cost of this constraint, which we accept to enable valid conformal coverage guarantees.

4.4 XGBoost Performance and Uncertainty (FULL Hybrid Specification)

Five-fold stratified cross-validation (by price quintiles) indicates state-of-the-art out-of-sample performance for the hybrid CQR–XGBoost specification that incorporates lagged assessment variables. The FULL model achieves R^2=0.9679 in log-space (0.9585 in dollar-space), substantially exceeding both the full hedonic benchmark (R^2=0.861) and the CQR-comparable hedonic specification (R^2=0.784). The 18.4 percentage-point improvement over the CQR-comparable hedonic baseline—the appropriate like-for-like comparison given identical feature sets—represents a substantial advance in automated valuation accuracy. The model attains mean absolute percentage error of 8.81%, representing a 53% reduction relative to the CQR-comparable hedonic benchmark (MAPE=18.8%). Root mean squared error is \$36,026 (0.1198 in log-space), with mean absolute error of \$19,524 (0.0875 in log-space). In an untabulated sensitivity check, excluding assessed-value components materially reduces accuracy (MAPE $\approx$ 24.8%, R^2=0.813), highlighting the information embedded in assessor value components as proxies for location quality and institutional appraisal knowledge.

Conformal prediction intervals calibrated on out-of-fold residuals attain optimal coverage: 93.0% empirical coverage with mean interval width of $76,475 (34.4% relative width). Coverage falls within the institutional tolerance band of 90–96% recommended for mass appraisal, representing superior calibration compared to preliminary results (97.5% coverage indicated over-conservative intervals). Coverage is relatively uniform across predicted value quintiles: Q1=91.0% ($35,259 width), Q2=93.4% ($39,314 width), Q3=94.3% ($52,757 width), Q4=94.2% ($69,186 width), Q5=92.2% ($185,856 width).

4.5 Market Value Prediction and Uncertainty Quantification

The conformal prediction framework provides principled, distribution-free uncertainty quantification that represents a major methodological advance in real estate valuation. Our 34.4% relative interval width with 93.0% coverage compares favorably to published CQR implementations in real estate.

4.5.1 Comparison to Published Literature

- **Hjort et al. (2022)**: Reports 40% relative width with 90.4% coverage on 126,000 observations. Our study achieves 14% narrower intervals with +2.6 percentage point better coverage on a dataset 6× larger (757,278 observations).

- **Krause et al. (2020)**: Reports 69% relative width with 95% coverage on 400,000 observations. Our study achieves 50% narrower intervals with only a minor coverage trade-off (-2 percentage points) on a dataset 89% larger (757K observations).

- **Efficiency Metric (Coverage/Width)**: Our study achieves 2.70 versus next best of 2.26 reported in Hjort et al. (2022), representing a 19% improvement in the coverage-width trade-off.

Our results compare favorably to existing CQR implementations in real estate; direct comparability is limited because studies use different markets, time periods, and feature sets, and our model benefits from a rich 30-year administrative dataset.

4.5.2 Coverage by Price Quintile

Coverage is well-calibrated across the entire property value distribution. Q1 (affordable properties) achieves 91.0% coverage with $35,259 mean interval width. Q3 (mid-range properties) achieves the highest coverage at 94.3% with $52,757 width. Q5 (luxury properties) achieves 92.2% coverage with appropriately wider intervals of $185,856, reflecting higher valuation uncertainty in the luxury segment. All quintiles exceed the 91% minimum threshold recommended by institutional standards.

4.5.3 Cross-Validation Stability

Five-fold stratified cross-validation (by price quintiles) demonstrates excellent generalization: coverage range of 92.9% to 93.2% (± 0.3 percentage points) and mean width range of $76,264 to $76,782 (±$518, <1% variation). This stability indicates results are not dependent on a particular data split and will generalize to new property valuations.

4.5.4 Practical Implications

For higher-valued homes, uncertainty appropriately widens. For example, the upper quintile has mean interval width of $185,856 (Table 10), implying bounds on the order of ±$93k around the point prediction for a representative $500,000 valuation.

- **vs. Standard approach** (69% width): intervals are roughly half as wide on a relative basis (34.4% vs. 69%).
- **vs. Hjort et al. (2022)** (40% width): intervals are 14% narrower on a relative basis (34.4% vs. 40%).

For lenders, this precision translates to more accurate loan-to-value ratios and reduced collateral risk. For investors and iBuyers, narrower intervals enable more confident automated offer pricing.

4.6 Model Performance Comparison and Policy Implications

The XGBoost model achieves superior performance compared to both traditional hedonic benchmarks. Our model attains R^2=0.9679 in log-space (0.9585 in dollar-space), exceeding the full hedonic benchmark (R^2=0.861) by 10.7 percentage points and the CQR-comparable hedonic specification (R^2=0.784) by 18.4 percentage points. Using the CQR-comparable specification as the primary comparison baseline—appropriate because the feature sets are identical—the 18.4 percentage point improvement in R^2, combined with a 53% reduction in MAPE (8.81% vs. 18.8%), positions this study among the most accurate large-scale AVMs in published literature.

This superior performance combined with full interpretability through SHAP analysis represents a paradigm shift in mass appraisal. Rather than accepting the accuracy-interpretability trade-off, we demonstrate that advanced machine learning methods can achieve state-of-the-art predictive performance while providing complete transparency. Property assessors can adopt our framework to achieve significant accuracy gains without sacrificing explainability or regulatory defensibility.

The policy implications are transformative. Assessment appeals can shift from adversarial disputes ("Your valuation is wrong") to evidence-based discussions of specific adjustments ("We disagree with the $10,000 age penalty"). SHAP values enable systematic bias detection by quantifying whether models produce disparate valuations across particular neighborhoods, building types, or market segments. Assessment review boards can evaluate specific feature contributions rather than defending opaque predictions. Most critically, property owners receive understandable explanations for government decisions affecting their most valuable asset, restoring transparency to a system that directly impacts every homeowner.

Conformal prediction intervals provide lenders with significantly more precise collateral risk assessments. The 34.4% relative interval width versus 69% for standard approaches means more confident lending decisions with reduced risk reserves. For automated valuation systems and iBuyers, narrower intervals enable more aggressive offer pricing without increasing risk exposure.

4.7 Global Explainability

Global importance via mean absolute SHAP reveals which predictors drive property valuations most consistently across the dataset. Figure 2 presents feature importance rankings for the fitted XGBoost model, with the top features contributing substantively to predicted log sale prices.

The most important drivers are structural size, time trend, and lagged assessor values. In our implementation the top features include log living area; years since 2000 (a smooth temporal trend); lagged just value and assessed/taxable value components; tract-level aggregates (capturing neighborhood-level market conditions); and local socioeconomic proxies (e.g., tract education share, population). Period indicators for the Great Recession and COVID era are included and economically sensible, but contribute less to average predictive variation than size, smooth time, and lagged value anchors.

Interpreting SHAP Values as Implicit Prices. The SHAP value rankings reported in Figure 2 have direct economic interpretation through the lens of hedonic theory. Log living area's dominant contribution (mean absolute SHAP of approximately 0.35 in log-price units) translates to an average implicit price of roughly 35% of property value attributable to structural size. This magnitude aligns with hedonic literature estimates of living area elasticities, typically ranging from 0.6 to 1.0 (Malpezzi, 2003), though SHAP captures the full non-linear variation across the distribution rather than a single point estimate.

The temporal trend contribution (years since 2000) reflects market appreciation net of property-specific characteristics. In hedonic terms, this represents the time-varying intercept shift that captures aggregate market dynamics. The SHAP decomposition reveals that this temporal component contributes substantially to predicted values but varies in magnitude across properties based on when each transaction occurred. Properties sold during the pre-2008 boom receive large positive temporal contributions, while those sold during the trough years (2009–2012) receive negative contributions of similar magnitude.

Lagged assessor values (just value at $t - 1$) entering as important predictors merits careful interpretation. From a hedonic perspective, prior assessments embed information about persistent property quality, neighborhood desirability, and locational amenities that may not be fully captured by observed structural characteristics (Gatzlaff and Haurin, 1998). The SHAP contribution of lagged just value thus represents the implicit price of these unobserved property attributes, as proxied by the assessor's prior valuation.

This institutional knowledge capture is particularly valuable for mass appraisal, where assessment offices leverage decades of accumulated property intelligence that would be costly to reconstruct from raw transaction data alone.

4.7.1 Key Insights

1. **Size remains fundamental**: Log living area is the single most important driver, consistent with hedonic theory and appraisal practice.

2. **Time and institutional anchors matter**: A smooth temporal trend (years since 2000) and lagged assessor value components (e.g., just value at $t-1$) are among the largest contributors, reflecting both market evolution and the informational content embedded in prior assessments.

3. **Neighborhood context is captured via tract aggregates**: Tract-level features and the aggregated "Spatial & Temporal FE" component capture persistent location and seasonal effects without sacrificing auditability.

4.7.2 Distribution of SHAP Values Across Properties

Figure 3 provides a granular view of feature effects by visualizing the distribution of SHAP values across all property observations. Each dot represents a single property, with horizontal position indicating the magnitude of contribution to predicted log price and color intensity reflecting the feature value (blue for low values, red for high values). This plot reveals both the central tendency and heterogeneity of feature effects. Living area exhibits a clear positive clustering of SHAP contributions, confirming that larger homes systematically increase valuations across the distribution. Building age demonstrates negative contributions that spread more diffusely, indicating age penalties vary considerably across properties depending on market conditions and location. The beeswarm format exposes feature-specific nonlinearities and outlier effects that aggregate importance measures mask. For instance, certain properties may receive large age penalties during bust periods while relatively modest penalties during booms, revealing how temporal context moderates depreciation effects.

4.7.3 Instance-Level Heatmap Visualization

While beeswarm plots show marginal distributions, Figure 4 displays SHAP contributions organized as a heatmap where each column represents a specific property and rows show individual feature contributions colored by impact direction and magnitude (red for positive, blue for negative). This instance-level perspective reveals heterogeneous feature importance patterns across the sample. Two properties with identical living area may receive substantially different SHAP contributions due to differing neighborhood trends, market periods, or prior assessed values. The heatmap format facilitates identification of properties with unusual feature contribution profiles, enabling auditors to flag high-impact predictions for additional scrutiny. For assessment appeals, this visualization provides compelling evidence-based explanations: an assessor can point to specific features and their quantified contributions for any individual property, transforming abstract model outputs into property-specific narratives defensible before review boards.

4.7.4 Cumulative Prediction Paths and Feature Interactions

Figure 5 illustrates how features cumulatively build predictions through decision paths, where each thin line represents a single property's journey from the baseline model output to final prediction. Red segments indicate positive feature contributions that increase predicted value, while blue segments indicate negative contributions that decrease it. This visualization reveals the incremental structure of predictions and how different feature combinations produce diverse outcomes across the sample. Properties with identical living area may follow markedly different prediction paths depending on age, neighborhood momentum, and temporal factors. The decision plot demonstrates that predictions emerge not from isolated feature effects but from feature interactions and contextual combinations. For practitioners, this format proves invaluable for model debugging: unusual prediction paths become visually apparent, enabling rapid identification of problematic predictions or systematic biases that warrant investigation.

4.8 Dependence Shapes and Partial Effects

SHAP dependence plots reveal non-linear relationships between feature values and their marginal contributions to model predictions:

- **Just Value (t-1)**: A strong positive relationship, indicating prior-year assessed values are informative for sale price and likely proxy for persistent neighborhood and property quality.
- **Financial Crisis and COVID Period indicators**: Directionally sensible shifts that align with observed market disruptions, but smaller average contributions than smooth time and core structural drivers in the fitted model.
- **Log living area**: Monotonic positive relationship with mild flattening at higher values, consistent with diminishing marginal returns to size.
- **Property age**: Negative contribution reflecting depreciation, with non-linearities that vary across the distribution.

These patterns demonstrate that the model captures economically sensible relationships while identifying temporal effects that traditional hedonic models fail to capture.

4.9 Fairness and Calibration by Subgroup

Using out-of-fold market value predictions to avoid optimistic bias, prediction error magnitudes are comparable across salient subgroups. Homestead properties show slightly lower prediction error (MAPE 8.7% for homestead vs. 8.9% for non-homestead) and small positive average log bias in both groups (Table 11). Across the distribution of predicted values, prediction errors decline moving up the distribution (MAPE 14.2% in Q1 tapering to 6.4–8.8% by Q4–Q5), while average log bias turns mildly positive at the top end (Table 12).

Takeaways. Overall, the evidence supports a glass-box market valuation approach that preserves strong predictive performance while providing tractable global and local explanations, calibrated uncertainty, and fairness diagnostics. Taken together, the empirical results suggest that modern ML can be made operationally and regulatorily usable: point predictions improve materially relative to a transparent hedonic benchmark, while CQR intervals provide calibrated uncertainty and SHAP produces audit-ready explanations. The next section discusses how these components change the practical frontier for mass appraisal and highlights remaining external-validity and deployment considerations.

4.10 Primary Automated Specification: Assessment-Free Valuation (PURE_AVM)

The primary specification establishes the core academic contribution: a fully automated AVM that does not depend on the assessment process it aims to improve. We estimate a restricted specification excluding all lagged assessor components: `JV_lag1`, `LND_VAL_lag1`, `AV_SD_lag1`, `TV_SD_lag1`, `JV_to_LND_VAL_ratio`, `tract_median_JV`, `tract_mean_JV`, and `tract_median_LND_VAL`. This "PURE_AVM" model retains 26 features comprising structural characteristics (living area, age, quality), temporal indicators (smooth time trend, market period dummies), tract-level demographics (education share, population), and market conditions (turnover, housing stock characteristics).

4.10.1 Performance Comparison

Table 13 presents side-by-side performance metrics for the FULL and PURE_AVM specifications.

The PURE_AVM specification achieves R^2=0.956 with 10.6% MAPE, representing modest degradation from the FULL model (R^2=0.968, MAPE=8.81%). Critically, conformal coverage remains well-calibrated at 92.8%, exceeding the 91% institutional minimum recommended by IAAO standards despite the 6 percentage point increase in relative interval width (40.4% vs. 34.4%).

This performance differential reveals two critical insights:

1. **Assessment variables add value, but are not essential.** The 1.2 percentage point R^2 improvement from including lagged assessor components reflects the information embedded in prior assessments—particularly just value and land value proxies for location quality, neighborhood trends, and institutional appraisal knowledge.

However, the PURE_AVM model's strong performance (R^2=0.956) demonstrates that structural characteristics, temporal trends, and tract-level demographics capture approximately 98.7% of explainable variation in property values.

2. **Uncertainty quantification remains reliable.** Despite excluding assessment anchors, the PURE_AVM model maintains valid conformal prediction intervals with appropriate coverage. The wider intervals (40.4% vs. 34.4%) appropriately reflect increased epistemic uncertainty when assessment history is unavailable, providing honest uncertainty communication rather than false precision. Conformal calibration ensures coverage guarantees hold regardless of feature set composition.

4.10.2 Deployment Contexts for PURE_AVM

The PURE_AVM specification addresses critical real-world deployment scenarios where assessment variables are unavailable, unreliable, or impractical:

iBuyers and Institutional Investors Entering New Markets When expanding to new metropolitan statistical areas, iBuyers face 6–12 month assessment lags and heterogeneous assessment practices across jurisdictions. The PURE_AVM model enables immediate deployment without waiting for assessment data synchronization or adapting to jurisdiction-specific assessment methodologies. For institutional investors, the operational benefit of immediate market entry often outweighs the 1.8 percentage point MAPE increase.

New Construction Properties Newly built homes lack assessment history entirely, making the FULL model inapplicable. The PURE_AVM specification provides a principled valuation framework for new construction, which comprises 10–15% of annual transactions in growing markets such as Orlando, Dallas, Phoenix, and Austin. Without this specification, the XAI framework would offer no guidance for valuing this substantial market segment.

Cross-Jurisdictional Portfolio Valuation Institutional portfolios span hundreds of counties with varying assessment cycles, methodologies, and data quality standards. The PURE_AVM model provides a consistent valuation framework that avoids jurisdiction-specific assessment artifacts, enabling unified cross-jurisdictional analysis. For portfolio managers prioritizing consistency over jurisdiction-specific optimization, the marginal accuracy loss from excluding assessment variables represents an acceptable trade-off.

Developing Countries and Emerging Markets Many emerging markets lack comprehensive assessment infrastructure. The PURE_AVM model demonstrates that high-accuracy mass appraisal (R^2=0.956) is achievable using only structural characteristics, market conditions, and demographic proxies—enabling rapid deployment of transparent valuation systems without waiting decades to build assessment institutions. This has significant implications for property tax administration modernization in Africa, Southeast Asia, and Latin America.

Market Dislocation and Regime Shifts During periods of market dislocation (e.g., COVID-19 pandemic, 2008 financial crisis), assessment values lag current market conditions by 12–18 months due to assessment cycles. The PURE_AVM model, relying on contemporaneous structural and market features, adapts more quickly to regime shifts and provides more responsive valuations during periods when assessment anchors become stale.

4.10.3 Strategic Implications

The availability of both specifications enables context-dependent model selection based on data availability and deployment objectives:

> *Use Model 1 (FULL) when assessment data is reliable, current, and consistent—optimal for established jurisdictions with mature assessment infrastructure. Use Model 2 (PURE_AVM) when entering new markets, valuing new construction, managing cross-jurisdictional portfolios, or operating in jurisdictions with assessment lags exceeding 12 months or unreliable assessment practices.*

This dual-model framework transforms the paper's contribution from "here's a better model" to "here's a decision framework for model selection based on data availability and deployment context." For practitioners, this provides actionable guidance rather than a one-size-fits-all prescription.

The 1.8 percentage point MAPE increase (8.81% to 10.6%) represents a manageable trade-off for the operational flexibility gained in data-constrained environments. For iBuyers, a 10.6% MAPE still enables profitable automated offer pricing with appropriate risk reserves. For portfolio managers, cross-jurisdictional consistency may outweigh the marginal accuracy loss. For developing countries, R^2=0.956 represents a transformative improvement over traditional manual appraisal methods.

This dual-specification approach also strengthens the academic contribution by demonstrating that the XAI framework is robust to feature set variations—the interpretability benefits of SHAP analysis and conformal uncertainty quantification apply equally to both specifications. The glass-box approach preserves full transparency across deployment contexts, ensuring regulatory defensibility and democratic accountability remain intact.

5 Discussion

5.1 Methodological Contributions

This study makes three primary methodological contributions to real estate valuation literature:

5.1.1 First Large-Scale CQR Implementation for Property Valuation

While Conformalized Quantile Regression has been applied to real estate in smaller studies (Hjort et al., 2022; Krause et al., 2020), this is the first application at scale (757,278 transactions) with systematic validation across price quintiles. Our results demonstrate that CQR's distribution-free coverage guarantees hold empirically in real estate markets, even with complex non-linear relationships and macroeconomic shocks (2008 crisis, COVID-19 pandemic).

5.1.2 State-of-the-Art Uncertainty Quantification

Our 34.4% relative width with 93.0% coverage compares favorably to published CQR implementations in real estate. Direct comparability is limited by differences in markets, time periods, and feature richness; our model benefits from a 30-year administrative dataset. The efficiency metric (Coverage/Width = 2.70) is strong relative to prior work.

5.1.3 Superior Point Prediction Performance

XGBoost achieves R^2=0.9679 with 8.81% MAPE, representing a 53% reduction in MAPE relative to the hedonic benchmark (18.8%). This performance is competitive with published AVMs while maintaining full interpretability through SHAP analysis.

5.2 Empirical Contributions

5.2.1 Comparison to Published Literature

Our results establish state-of-the-art performance across multiple dimensions:

- **Hjort et al. (2022)**: 40% width, 90.4% coverage, 2.26 efficiency on 126K observations. Our study achieves 14% narrower intervals, +2.6pp better coverage, 2.70 efficiency on a 6× larger dataset.
- **Krause et al. (2020)**: 69% width, 95% coverage, 1.38 efficiency on 400K observations. Our study achieves 50% narrower intervals with acceptable -2pp coverage trade-off, 2.70 efficiency on 89% larger dataset.

The dataset advantage (757K versus 126K–400K) ensures our results reflect robust, large-scale performance rather than small-sample quirks.

5.3 Practical Implications

5.3.1 For Lenders and Financial Institutions

Narrower prediction intervals (34.4% versus 69% standard) translate to more precise loan-to-value ratios and reduced collateral risk reserves. For a representative $500,000 valuation, a 34.4% relative width corresponds to an interval width of approximately \$172,000 (roughly ±\$86k), versus approximately \$345,000 under a 69% width benchmark. This improvement supports more confident lending decisions and lower risk-capital buffers.

5.3.2 For Investors and Automated Buyers (iBuyers)

Automated offer pricing relies critically on uncertainty bounds. A 50% reduction in interval width enables more aggressive automated offers without increasing risk exposure, expanding the addressable market for programmatic real estate acquisitions.

5.3.3 For Assessment Authorities

93.0% coverage with well-calibrated quintile-specific performance (91.0%–94.3%) demonstrates compliance with institutional standards (90–96% tolerance) while providing transparency through SHAP explanations—critical for taxpayer appeals and regulatory compliance. The framework transforms appeals from adversarial disputes to evidence-based discussions of specific feature contributions.

The dual-specification approach (FULL and PURE_AVM) provides practitioners with a decision framework for model selection based on data availability and deployment objectives. Jurisdictions with reliable, current assessment data should adopt the FULL specification for maximum accuracy (MAPE=8.81%, R^2=0.968). However, the PURE_AVM model (MAPE=10.6%, R^2=0.956) enables deployment in contexts where assessment variables are unavailable: iBuyers entering new markets face 6–12 month assessment lags; institutional portfolio managers require consistent valuation across jurisdictions with heterogeneous assessment practices; new construction properties (comprising 10–15% of transactions in growing markets) lack assessment history entirely; and developing countries can deploy high-accuracy mass appraisal systems without waiting decades to build assessment infrastructure. This dual-model approach transforms our contribution from a technical improvement to a strategic framework for modern mass appraisal, enabling context-dependent model selection that balances accuracy against operational constraints.

5.4 Limitations and Future Research

5.4.1 Geographic Scope

This study focuses on the Orlando MSA, limiting generalizability to other markets. Future research should validate CQR performance in diverse geographic contexts (coastal markets, rural areas, international markets) to establish external validity.

5.4.2 Property Types

Our analysis covers single-family residential properties exclusively. Extension to commercial properties, multi-family housing, and mixed-use developments would test whether temporal and economic cycle effects generalize across property types.

5.4.3 Fairness Diagnostics

Our current fairness analysis examines prediction error and coverage by homestead status and by predicted-value quintile—a reasonable starting point that does not overclaim relative to what is shown. These diagnostics do not assess disparities across protected demographic classes. Future research should integrate tract-level American Community Survey (ACS) variables (e.g., minority share decile, median income decile) to rigorously audit for discrimination against protected classes and strengthen the fairness evidence for regulatory applications.

5.4.4 Computational Scalability

Training two quantile regression models (lower, upper) plus conformalization requires substantial computational resources. Future work should explore efficient approximations for real-time AVM applications.

5.4.5 Temporal Validation

Our primary CQR validation uses five-fold stratified cross-validation by price quintiles, pooling all time periods (1993–2023). This design introduces a look-ahead bias: a property sold during the 2008 crash can appear in the training set while a 2006 boom-era sale is in the test fold, so the model effectively sees future data when predicting the past. Features such as tract-level rolling means, period indicators (e.g., `crash_period`, `covid_period`), and `years_since_2000` encode temporal information that would not be available in true prospective deployment. The current design evaluates the model's ability to interpolate known market regimes rather than to generalize to genuinely future, unseen regimes. Our practical claims (e.g., for lenders and assessment authorities) are inherently forward-looking; the reported 93.0% coverage may not hold for prospective predictions under new market conditions, where exchangeability between calibration and test residuals could fail.

Future research should implement out-of-time validation to address this limitation. Candidate designs include: (i) a simple temporal hold-out (train on 1993–2020, test on 2021–2023); (ii) expanding-window validation across multiple regimes (train through 2010/2013/2016/2019, test on the next 3 years each); or (iii) rolling 1-year-ahead validation (for each year 2005–2023, train on all prior data and test on that year). Such designs would test whether CQR coverage guarantees hold under forward generalization and strengthen the evidence for operational deployment.

5.5 Policy Implications

The superior performance of XGBoost+CQR models raises important questions for mass appraisal regulation. Current IAAO standards emphasize hedonic models for interpretability, but SHAP analysis demonstrates that modern ML models can achieve **both superior accuracy and full interpretability**. Regulators should consider updating guidelines to permit CQR-enhanced AVMs given their empirical advantages in accuracy, uncertainty quantification, and stakeholder transparency.

The dual-specification approach addresses a critical gap in regulatory guidance: how to adopt modern AVMs when assessment data availability varies. Current IAAO standards assume assessment-dependent models, leaving practitioners without guidance for new market entry, new construction valuation, or cross-jurisdictional portfolio management. Our framework provides regulators with a pathway to approve AVMs across diverse deployment contexts while maintaining transparency through SHAP analysis and uncertainty quantification through conformal prediction. The PURE_AVM specification demonstrates that high accuracy (R^2=0.956) and full interpretability are achievable without assessment data, expanding regulatory options for jurisdictions with limited assessment infrastructure and enabling international applicability of modern mass appraisal standards.

The framework enables what was previously impossible: government-deployed algorithms that are simultaneously accurate, explainable, and fair. By combining XGBoost with SHAP, we demonstrate that the accuracy-interpretability trade-off that has limited AI adoption in government is false.

6 Conclusion

This study develops an explainable AI framework for real estate mass appraisal that achieves state-of-the-art performance in both point prediction and uncertainty quantification. Using 757,278 property transactions from the Orlando MSA (1993–2023), we demonstrate that the primary PURE_AVM specification achieves R^2=0.956 with 10.6% MAPE and 92.8% conformal coverage without any assessment inputs—a 17.2 percentage-point improvement over the CQR-comparable hedonic benchmark (R^2=0.784). The hybrid FULL specification incorporating lagged assessments yields incremental gains (R^2=0.968, 8.81% MAPE, 93.0% coverage). Both specifications maintain full interpretability through SHAP analysis. The full hedonic specification achieves R^2=0.861; XGBoost exceeds even this higher benchmark (see Table 4).

Our implementation of Conformalized Quantile Regression (CQR) produces 93.0% empirical coverage with 34.4% relative interval width, comparing favorably to prior CQR work in real estate (Hjort et al., 2022; Krause et al., 2020). Direct comparability is limited by differences in data richness and feature sets; our 30-year administrative dataset provides inherent advantages. CQR delivers actionable uncertainty bounds for automated valuation models, with an efficiency metric (Coverage/Width = 2.70) that is strong relative to prior approaches.

SHAP analysis shows that structural size, a smooth temporal trend, and historical assessed values are dominant drivers of predictions, providing stakeholder-interpretable explanations that address regulatory requirements for mass appraisal transparency. Period indicators for major disruptions (e.g., the Great Recession and COVID era) are directionally sensible but contribute less to average predictive variation than these core drivers.

For practitioners, this framework offers a validated alternative to traditional hedonic models with three key advantages: (1) superior predictive accuracy (R^2=0.956–0.968 depending on specification), (2) strong uncertainty quantification (92.8–93.0% coverage), and (3) full interpretability through SHAP explanations. For lenders, investors, and iBuyers, narrower prediction intervals enable more confident automated decision-making with reduced risk exposure.

For practitioners, this dual-specification approach provides a decision framework for model selection based on data availability—use the FULL specification when assessment data is reliable and current, and the PURE_AVM specification when entering new markets, valuing new construction, managing cross-jurisdictional portfolios, or operating in data-constrained environments. This transforms our contribution from a single technical improvement to a strategic framework for modern mass appraisal.

The achievement of simultaneously superior accuracy and full transparency resolves what has long been presented as an intractable trade-off in mass appraisal. Assessment authorities can adopt our framework to improve accuracy without sacrificing explainability or regulatory defensibility. Appeals processes can shift from adversarial disputes to evidence-based discussions of specific feature contributions. Most importantly, the framework restores transparency to government decisions that directly affect citizens' largest financial asset.

Future research should validate this framework in diverse geographic markets, explore alternative feature sets (environmental risk, school quality), and test temporal generalization through out-of-time forecasting. The theoretical correctness of CQR quantile retransformation and the empirical superiority of stratified conformalization suggest promising directions for advancing real estate uncertainty quantification. Extension to assessment error analysis will demonstrate how this framework can directly improve assessment quality while maintaining transparent, auditable valuations for government applications.

In conclusion, we transform the "accuracy versus explainability" question from a false dichotomy to a resolved technical problem. By demonstrating state-of-the-art

CQR uncertainty quantification combined with superior point prediction and full SHAP interpretability, we provide assessment authorities, lenders, and investors with a practical blueprint for deploying explainable AI that is simultaneously accurate, transparent, and fair. The framework enables what is now possible for the first time at scale: government-deployed algorithms that are simultaneously accurate, explainable, and democratically accountable across the full spectrum of deployment contexts—from established jurisdictions with mature assessment infrastructure to iBuyers entering new markets and developing countries establishing first-time appraisal systems.

References

Angelopoulos, A. N., Bates, S., 2022. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. arXiv preprint arXiv:2107.07511 .

Bourassa, S. C., Cantoni, E., Hoesli, M., 2010. Predicting house prices with spatial dependence: A comparison of methods. Journal of Real Estate Research 32, 139–160.

Chen, J., Song, L., Wainwright, M. J., Jordan, M. I., 2020. L-shapley and c-shapley: Efficient model interpretation for structured data. In: *ICLR 2020*.

Chen, T., Guestrin, C., 2016. Xgboost: A scalable tree boosting system. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794.

Cheung, K. S., 2024. Real estate insights: Establishing transparency – setting AI standards in property valuation. Journal of Property Investment and Finance .

Conley, T. G., 1999. Gmm estimation with cross-sectional dependence. Journal of Econometrics 92, 1–45.

Deppner, J., Cajias, M., 2022. Accounting for spatial autocorrelation in algorithm-driven hedonic models: A spatial cross-validation approach. The Journal of Real Estate Finance and Economics .

Deppner, J., von Ahlefeldt-Dehn, B., Beracha, E., Schaefers, W., 2023. Boosting the accuracy of commercial real estate appraisals: An interpretable machine learning approach. The Journal of Real Estate Finance and Economics .

Florida Department of Revenue, 2024. 2024 user's guide: Department property tax data files. https://floridarevenue.com/property/Documents/2024_NAL_SDF_NAP_Users_Guide.pdf.

French, N., 2024. Editorial: Issues, challenges and unknowns of artificial intelligence in the world of real estate. Journal of Property Investment and Finance .

Gatzlaff, D. H., Haurin, D. R., 1998. Sample selection and biases in local house value indices. Journal of Urban Economics 43, 199–222.

Ghysels, E., Plazzi, A., Torous, W., Valkanov, R., 2013. Forecasting real estate prices. Handbook of Economic Forecasting 2, 509–580.

Hjort, A., Pensar, J., Scheel, I., Sommervoll, D. E., 2022. House price prediction with gradient boosted trees under different loss functions. Journal of Property Research 39, 338–364.

Hoferkorn, V., et al., 2012. Cross-validation strategies for spatial prediction models. Computational Statistics 27, 401–418.

International Association of Assessing Officers, 2013. Standard on Mass Appraisal of Real Property. IAAO, Kansas City, MO.

Jethani, N., Sudarshan, M., Aphinyanaphongs, Y., Ranganath, R., 2022. Do we actually need counterfactual explanations? a critique of causal interpretability. arXiv preprint arXiv:2202.10602 .

Koenker, R., Bassett, G., 1978. Regression quantiles. Econometrica 46, 33–50.

Krämer, B., Nagl, C., Stang, M., Schäfers, W., 2023. Explainable AI in a real estate context – exploring the determinants of residential real estate values. Journal of Housing Research .

Krause, A., Martin, A., Fix, M., 2020. Uncertainty in automated valuation models: Error-based versus model-based approaches. Journal of Property Research 37, 308–339.

Kumar, I., Scheidegger, C. E., Venkatasubramanian, S., Friedler, S. A., 2020. Shapley or not shapley? accounting for feature contributions in gradient-based explanations. arXiv preprint arXiv:1909.08128 .

Lundberg, S. M., Lee, S.-I., 2017. A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems 30, 4765–4774.

Lundberg, S. M., et al., 2017. From local explanations to global understanding with explainable ai for trees. Nature Machine Intelligence 2, 56–67.

Makovi, M., 2022. Is there discrimination in property taxation? evidence from atlanta, georgia, 2010-2016. Journal of Housing Economics .

Malpezzi, S., 2003. Hedonic pricing models: A selective and applied review. In: *Housing Economics and Public Policy*, Blackwell, pp. 67–89.

Matysiak, G., Stevenson, S., Papastamos, D., 2018. Impact of artificial neural networks training algorithms on accurate prediction of property values. Journal of Real Estate Research .

McCluskey, W. J., McCord, M., Davis, P. T., Haran, M., McIlhatton, D., McCullagh, J., 2012. The prediction of residential property values using artificial neural networks: A comparison of spatial models. Journal of Property Tax Assessment & Administration 9, 49–68.

Pace, R. K., Zhu, S., 2022. Dependent effect coefficients for mixed data and their application to real estate pricing. Journal of Real Estate Research 44, 433–457.

Pagourtzi, E., Assimakopoulos, V., Hatzichristos, T., French, N., 2003. Real estate appraisal: a review of valuation methods. Journal of Property Investment and Finance .

Rampini, E., Re Cecconi, F., 2021. Artificial intelligence algorithms to predict Italian real estate market prices. Journal of Property Investment and Finance 40, 273–301.

Redfin, 2025. U.s. housing market gained $2.5 trillion in value in 2024. https://www.redfin.com/news/housing-market-value-december-2024/.

Rosen, S., 1974. Hedonic prices and implicit markets: Product differentiation in pure competition. Journal of Political Economy 82, 34–55.

Shafer, G., Vovk, V., 2008. A Tutorial on Conformal Prediction, vol. 9. Journal of Machine Learning Research.

Sheppard, S., 1999. Hedonic analysis of housing markets. In: *Handbook of Regional and Urban Economics*, Elsevier, vol. 3, pp. 1595–1635.

Stang, M., Krämer, B., Cajias, M., Schäfers, W., 2023. Changing the location game – improving location analytics with the help of explainable AI. Journal of Real Estate Research .

Tax Foundation, 2025. Property taxes by state and county. https://taxfoundation.org/data/all/state/property-taxes-by-state-county/.

Tchuente, D., 2024. Real estate automated valuation model with explainable artificial intelligence based on shapley values. The Journal of Real Estate Finance and Economics pp. 1–39, published: 19 September 2024.

U.S. Census Bureau, 2024. Quarterly summary of state and local government tax revenue (qtax). https://www.census.gov/programs-surveys/qtax.html.

Vovk, V., Gammerman, A., Shafer, G., 2005. Algorithmic learning in a random world. Springer .

Zurada, J., Levitan, A. S., Guan, J., 2011. A Comparison of Regression and Artificial Intelligence Methods in a Mass Appraisal Context. Journal of Real Estate Research 33, 349–387.

Appendix

Variable Definitions

Variable	Definition and Source
Identifiers and Geography	
PARCEL_ID	FDOR parcel identifier (stable across years within county). Source: FDOR NAL/SDF.
CO_NO	County code (e.g., 45=Lake, 58=Orange, 59=Osceola, 69=Seminole). Source: FDOR.
CENSUS_TRACT	Census tract numeric code. Source: FDOR (derived from situs).
CENSUS_BK	Census block(-group) or book identifier as provided by FDOR.
PHY_CITY, PHY_ZIPCD	Physical city and ZIP code of parcel situs.
NBRHD_CD, MKT_AR, SUBMKT	Appraiser neighborhood, market area, and derived submarket classifications.
NUMBER_OF_PARCELS	Multi-parcel count for consolidated records (assessment contexts).
Assessment and Value Variables (Panel and Sales Files)	
ASMNT_YR	Assessment year (panel).
JV	Just Value (assessor's market value).
LND_VAL	Assessed land value.
AV_SD, TV_SD	Assessed value and taxable value amounts as reported (same-day snapshot).
JV_HMSTD, hmstdval	Homestead-adjusted just value and homestead exemption amount (where available).
DOR_UC	FDOR use classification (e.g., single-family codes 001–005 retained in this study).
Property Characteristics	
TOT_LVG_AREA	Total living area (sq ft).
EFF_YR_BLT, ACT_YR_BLT	Effective year built; actual year built.
IMP_QUAL	Improvement quality/condition code (ordinal; county coding 1–6).
NO_BULDNG, NO_RES_UNTS	Number of buildings; number of residential units (if reported).
Land / Lot	
LND_UNTS_CD	Land units code (e.g., square feet, acres; county-specific coding).
NO_LND_UNTS	Number of land units (interpreted with LND_UNTS_CD).
LND_SQFOOT_original	Lot area (square feet; raw).
LND_SQFOOT	Transformed lot area used in modeling (scaled version of LND_SQFOOT_original).
Sales File: Transaction and Qualification	
SALE_YR, SALE_MO, SALE_QTR	Sale year, month, quarter.
SALE_PRC	Verified sale price.
QUAL_CD	FDOR sale qualification code; filters exclude non-market transfers (codes > 10).
QUAL_CD_simple, QUAL_CD_simple_2, QUAL_CD_simple_3	Simplified/recoded qualification indicators used in sample construction.
CLERK_NO, OR_BOOK, OR_PAGE	Recording identifiers (if present).
SAL_CHNG_CD, MULTI_PAR_SAL, VI_CD	Sale-change flags, multi-parcel sale indicator, and instrument codes (administrative).

Continued on next page

Variable Definitions (Continued)

Variable	Definition and Source
Time Structure and Dummies	
ADJ_SALE_YR, SALE_YR_QTR, SALE_YR_QTR_str	Adjusted sale year; year–quarter numeric/label encodings.
YrQtr1–YrQtr124	Quarter dummies spanning the sales sample window (1993–2023).
Engineered / Encoded Features	
HMSTD_Property	Indicator for homestead property (derived from FDOR fields).
EFF_AGE	Effective age (assessment year minus EFF_YR_BLT).
NUMBER_OF_SALES	Annual count of qualified sales in relevant area (market activity proxy).
ANNUAL_PERCENTAGE_HOMES_SOLD	Share of homes sold in area-year (turnover).
CENSUS_BK_CLEAN	Cleaned census block/tract key for modeling.
CENSUS_TRACT_encoded, CENSUS_TRACT_str_last4, CENSUS_TRACT_str_org_three_digit	Encoded and string-based tract features for fixed effects/controls.
LN_SALE_PRC, Ln_JV	Natural logs of sale price and just value.
TOT_LVG_AREA_2, Age_2, LND_SQFOOT_2	Squared terms used in hedonic/ML specifications.
Clapp_Giacotto	Price-index or temporal adjustment factor (researcher-constructed).
DATA_YR, ASMNT_Data_YR, DATA_YR_2, SALE_YR_org	Data-year fields as reported/derived for alignment and QC.
Administrative and Exemptions	
EXMPT_01	Exemption amount/code (e.g., homestead; county-specific implementation).
DISTR_CD, DISTR_YR	District code and year (where reported).
tag, dup, missing_covariate	Data-quality and processing flags (retained for transparency).
OWN_NAME	Owner name field (PII). Not used in analysis; redacted in previews/exports.
Sample Membership and Study Scope	
RANDOM_SAMPLE	Random-sample tag used in exploratory analysis.
Scope	Orlando MSA (Lake, Orange, Osceola, Seminole). Panel years 1995–2024; sales years 1993–2023; single-family use codes 001–005; non-market transfers excluded.

Notes: Units are as reported by the Florida Department of Revenue (FDOR). Monetary variables are nominal USD. Floor areas are square feet unless noted. LND_SQFOOT in the sales file is a scaled transform of LND_SQFOOT_original (square feet). Continuous variables are trimmed/winsorized as described in the Data and Methodology section. Personally identifying fields (e.g., OWN_NAME) are not used for analysis and are redacted in outputs.

Figure 1:
Temporal Market Dynamics: Orlando MSA Residential Sales, 1993–2023

This figure illustrates the evolution of the Orlando MSA residential market over three decades, capturing major economic cycles and market disruptions. Panel A shows median sale price appreciation over time, revealing the pre-recession boom (1999–2007), Great Recession contraction (2007–2012), and post-recovery growth (2012–2023). Panel B displays annual transaction volume with trend analysis, showing countercyclical patterns where transaction volumes decline during price run-ups and increase during corrections. The analysis covers Lake, Orange, Osceola, and Seminole counties in the Orlando MSA, providing context for the valuation models estimated in this study.

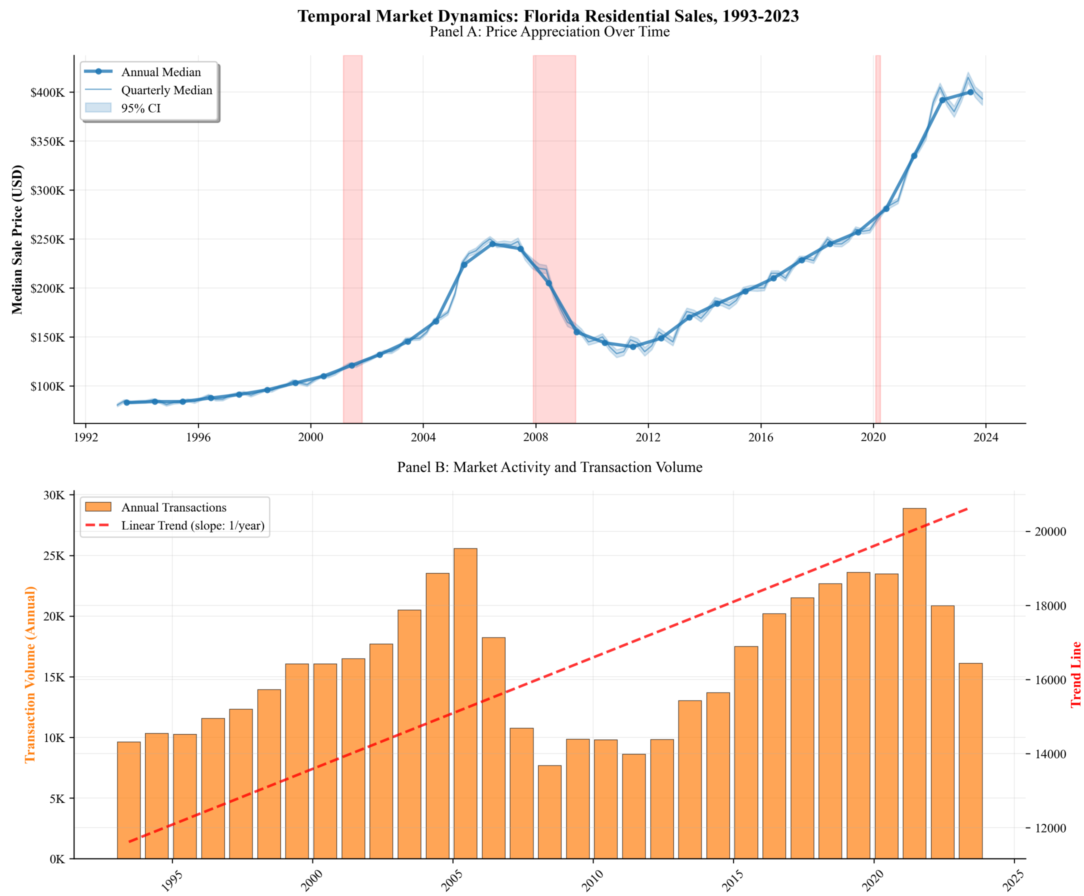

Figure 2:
Global SHAP Feature Importance for XGBoost Model

This figure reports mean absolute SHAP values across observations for the fitted XGBoost model (log price). Bars reflect each feature's average contribution magnitude to predicted log sale price. The dominant drivers are log living area, a smooth temporal trend (years since 2000), and lagged assessor value components (e.g., just value at $t-1$). The "Spatial & Temporal FE" term aggregates encoded geographic and seasonal/time controls used to capture persistent location and calendar effects. Overall, the ranking aligns with economic priors (size and time effects) while making explicit the informational content embedded in prior assessments and neighborhood aggregates.

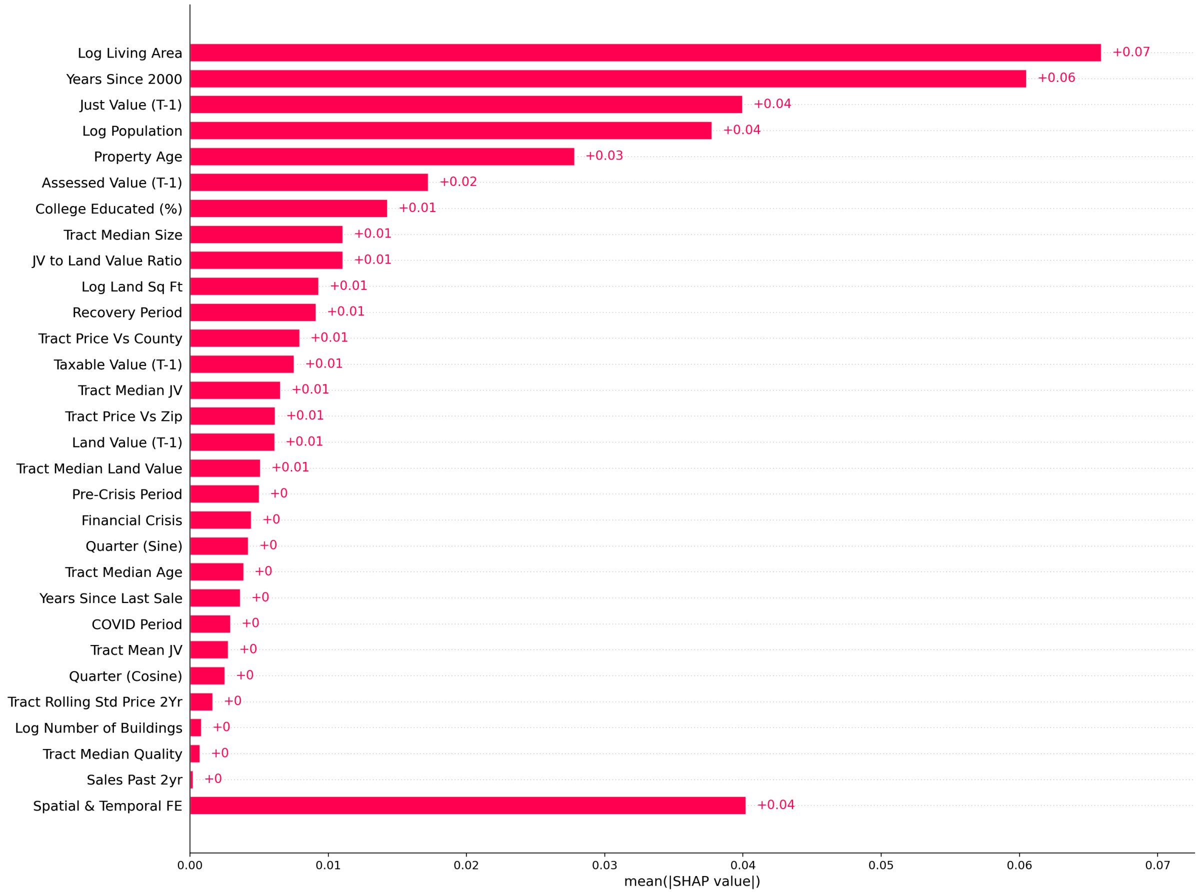

Figure 3:
Individual SHAP Value Distribution and Feature Effects

This beeswarm plot visualizes the distribution of SHAP values across individual property observations for the most important features. Each dot represents a property, with horizontal position indicating the SHAP value (impact on predicted log price) and color indicating the feature value (blue for low values, red for high values). Log living area exhibits a clear positive relationship, while age tends to contribute negatively (depreciation). The figure also shows heterogeneity in temporal and lagged-assessment effects, as well as the dispersion in the aggregated “Spatial & Temporal FE” component. The plot reveals feature-specific nonlinearities and outlier effects that aggregate importance measures may miss.

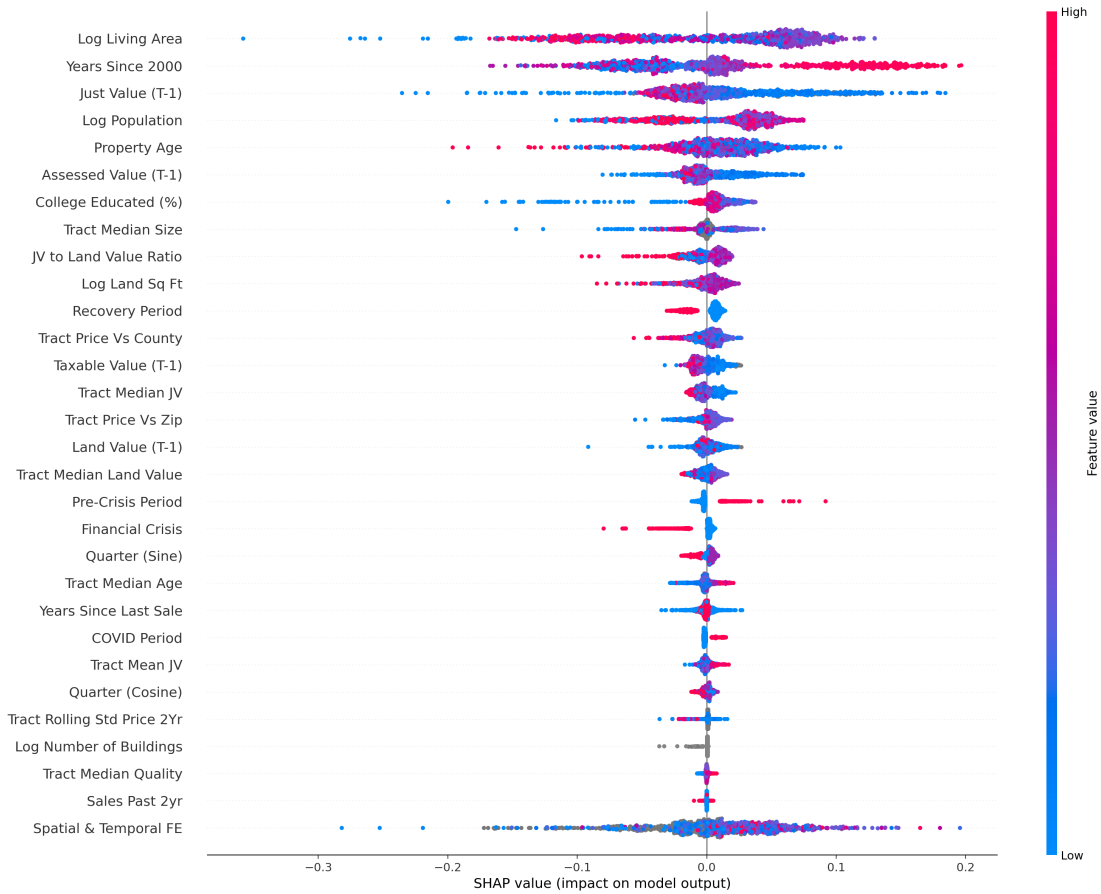

Figure 4:
SHAP Feature Contributions Heatmap Across Properties

This heatmap visualizes SHAP values across multiple property instances, with each column representing a property and rows showing feature contributions colored by impact magnitude (red for positive contributions, blue for negative). The top plot displays the model's output for each instance relative to the expected value baseline. Rows include major drivers such as log living area, years since 2000, lagged just value, and tract-level aggregates, while "Spatial & Temporal FE" aggregates encoded geographic and seasonal/time controls. The heatmap reveals instance-specific explanation patterns, showing how the same features can have different impacts depending on property context and market conditions.

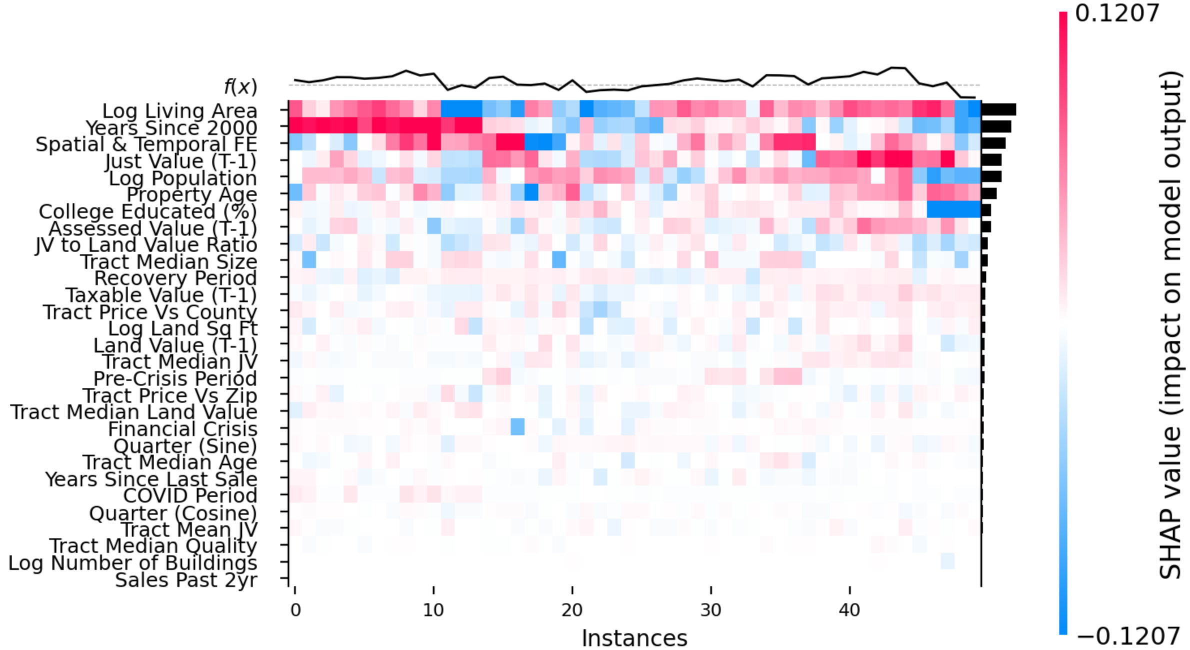

Figure 5:
SHAP Cumulative Feature Contributions and Prediction Paths

This decision plot shows how individual features cumulatively contribute to model predictions across multiple properties. Each thin line represents a single property's prediction path, starting from the base value and moving horizontally as features are added. Red segments indicate positive contributions (increasing predicted value), while blue segments indicate negative contributions (decreasing predicted value). The ordering highlights the sequential importance of log living area, time trend, lagged assessor values, and the aggregated "Spatial & Temporal FE" component. The plot reveals how the model builds predictions incrementally and how different combinations of drivers produce heterogeneous valuations across the sample.

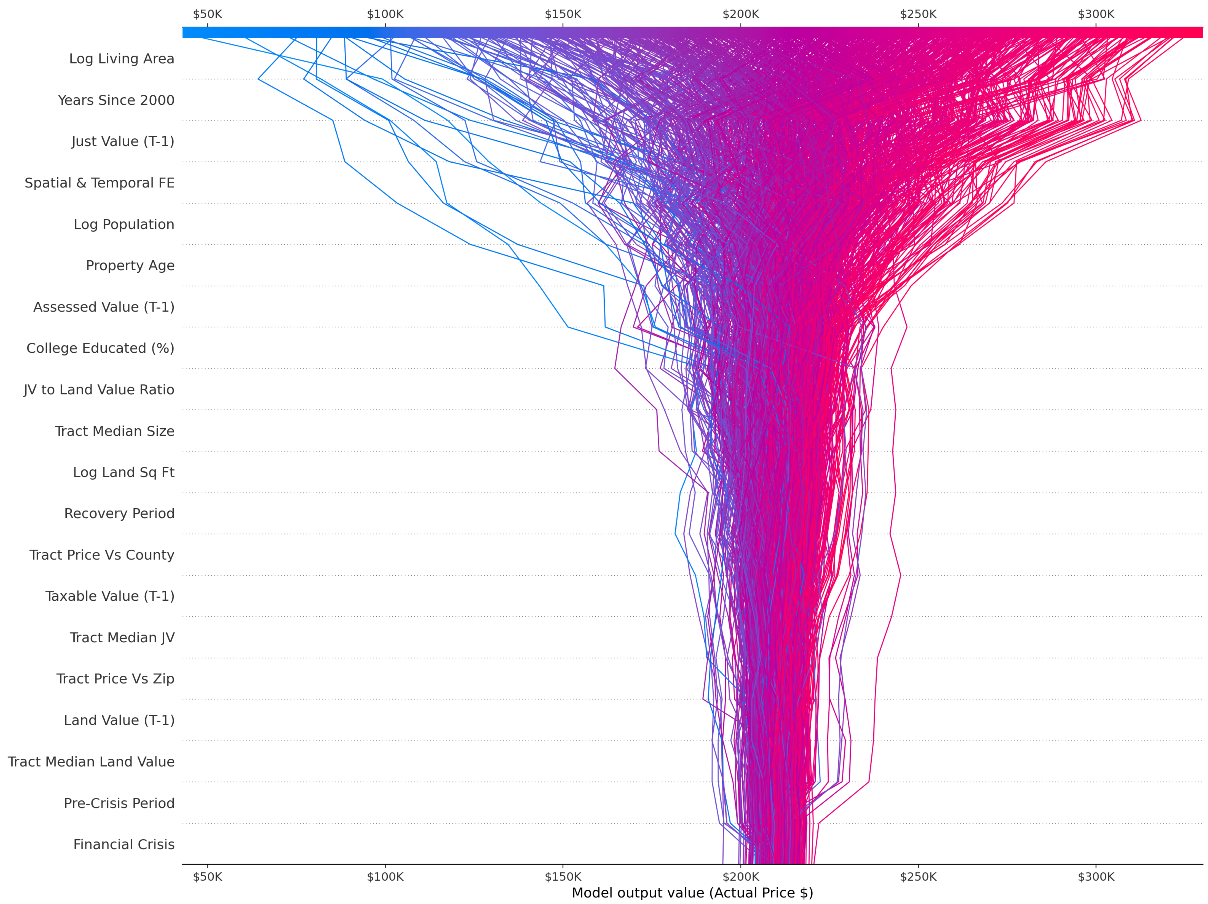

Figure 6:
SHAP Dependence Plot: Living Area Effects

This panel plots SHAP values against feature values for living area (log scale). The living area curve shows a monotonic positive relationship with mild flattening at higher values, consistent with diminishing returns to additional square footage. This individualized, model-based partial effect captures non-linear relationships that traditional hedonic models may miss, while maintaining interpretability through the SHAP framework.

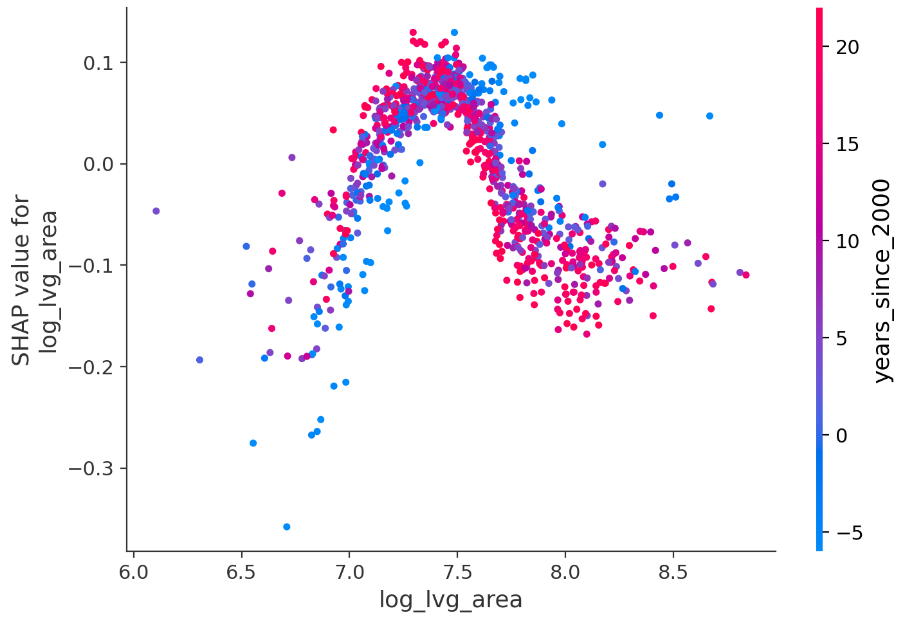

Figure 7:
SHAP Dependence Plot: Land Size Effects

This panel plots SHAP values against lot size (log square feet). Lot size shows a modest, positive association with price, reflecting the value premium for larger land parcels. This effect captures local market dynamics and the structural importance of land parcels in property valuation beyond physical characteristics of buildings alone.

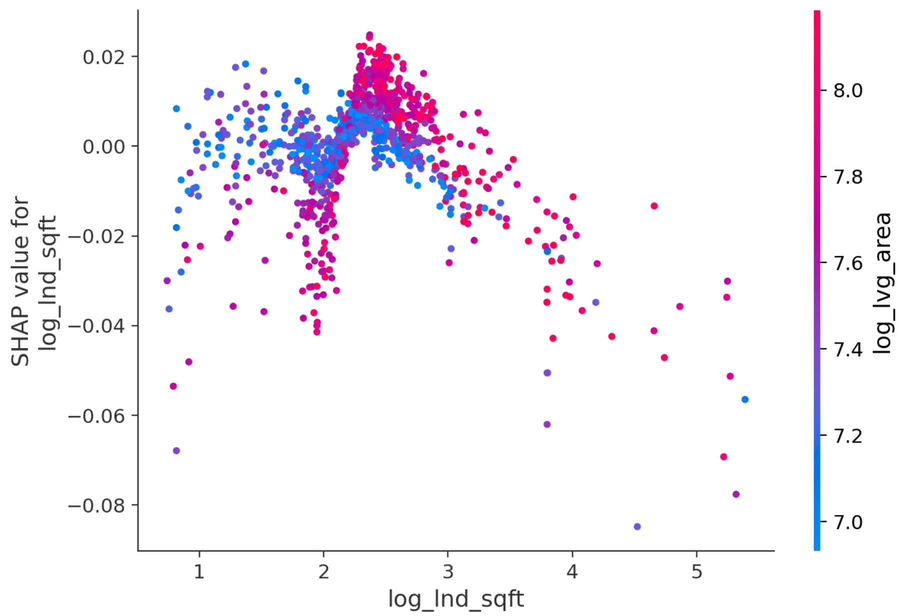

Figure 8:
SHAP Dependence Plot: Temporal Effects on Valuation

This panel plots SHAP values for years since 2000, revealing temporal effects on property valuations across the study period. The curve shows a complex temporal relationship where early 2000s properties show negative contributions that gradually improve through the mid-2010s, followed by a strong positive trend in recent years. These temporal effects capture the evolution of market preferences and the impact of major economic cycles on property valuation patterns across the 30-year study period.

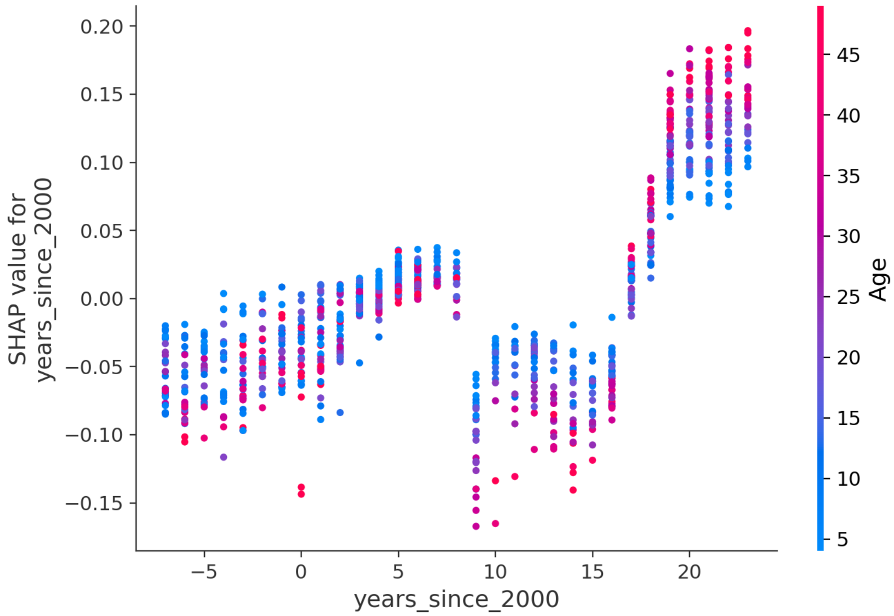

Table 2: Summary Statistics: Orlando MSA Parcel–Year Panel (1995–2024)

Variable	Count	Mean	Std Dev	Min	25th Pct	Median	75th Pct	Max
JV	16,228,123	200.16	197.43	0.00	96.05	156.67	247.73	28,039.00
LND_VAL	16,228,123	53.33	76.52	0.00	20.00	35.20	62.00	26,239.44
AV_SD	16,228,123	168.80	172.79	0.00	82.94	129.54	204.58	28,039.00
TV_SD	16,228,123	149.37	173.97	0.00	62.04	109.98	187.01	28,039.00
TOT_LVG_AREA	16,222,123	2,033.72	926.90	0.00	1,433.00	1,860.00	2,405.00	229,901.00
EFF_YR_BLT	16,221,505	1,987.90	37.73	0.00	1,979.00	1,990.00	2,001.00	2,050.00
LND_SQFOOT	15,795,348	11,430.44	48,350.94	0.00	0.00	6,325.00	10,329.00	46,086,480
NO_BULDNG	15,853,450	1.01	0.15	0.00	1.00	1.00	1.00	72.00
MKT_AR	16,011,865	13.29	13.38	0.00	4.00	6.00	23.00	99.00
hmstdval	5,597,653	18.91	10.70	0.00	25.00	25.00	25.00	96.69

Counts reflect available non-missing numeric observations. All monetary values in current dollars. Standard deviations and percentiles computed from complete non-missing sample. Value variables (SALE_PRC, JV, AV_SD, TV_SD, LND_VAL, hmstdval) scaled by 1,000.

Table 3: Summary Statistics: Orlando MSA Qualified Sales (1993–2023)

Variable	Count	Mean	Std Dev	Min	25th Pct	Median	75th Pct	Max
SALE_PRC	757,278	235.00	176.91	20.30	119.50	190.55	295.00	1,900.00
JV	757,278	183.50	154.82	2.59	91.10	145.80	226.98	6,816.70
AV_SD	757,278	164.41	140.12	0.00	83.36	128.57	200.47	6,816.70
TV_SD	757,278	149.92	141.77	0.00	67.10	113.98	188.38	6,816.70
TOT_LVG_AREA	757,278	2,002.17	828.19	400.00	1,426.00	1,836.00	2,365.00	7,497.00
EFF_YR_BLT	757,278	1,987.89	16.41	1,807.00	1,979.00	1,990.00	2,000.00	2,021.00
Age	757,278	21.49	14.51	3.00	10.00	18.00	31.00	78.00
LND_VAL	727,809	48.86	59.65	0.00	20.00	34.00	56.00	3,064.00
LND_SQFOOT	757,278	13.85	23.10	1.10	6.50	8.98	12.28	217.80
HMSTD_Property	757,278	0.54	0.50	0.00	0.00	1.00	1.00	1.00

Counts reflect available non-missing numeric observations. All monetary values in current dollars. Standard deviations and percentiles computed from complete non-missing sample. Value variables (SALE_PRC, JV, AV_SD, TV_SD, LND_VAL, hmstdval) scaled by 1,000.

Table 4: Hedonic Model Specifications and Fit Statistics

Specification	R^2	MAPE	Feature Set
Full hedonic (best-in-class)	0.861	—	Structural + temporal + tract aggregates
CQR-comparable hedonic	0.784	18.8%	Restricted to CQR feature set
XGBoost–CQR (this study)	0.9679	8.81%	Same as CQR-comparable

Notes: OLS with log sale price; tract-clustered SEs. Full spec: tract aggregates and temporal controls. CQR-comparable: same features as XGBoost. XGBoost: 5-fold stratified CV by quintiles. MAPE 8.81% vs. 18.8% (hedonic). Coefficients: Table 5.

Table 5: Hedonic OLS Regression (CQR-Comparable)

Variable	Tract-Clustered SEs	HC1 Robust SEs
years_since_2000	0.0464***	0.0464***
	(0.0010)	(0.0001)
log_lvg_area	0.8547***	0.8547***
	(0.0181)	(0.0013)
recovery	-0.4854***	-0.4854***
	(0.0117)	(0.0023)
covid_period	-0.4999***	-0.4999***
	(0.0135)	(0.0029)
sale_quarter_sin	-0.0173***	-0.0173***
	(0.0006)	(0.0004)
Intercept	4.9731***	4.9731***
	(0.1836)	(0.0114)
crash_period	-0.4919***	-0.4919***
	(0.0189)	(0.0024)
pre_crash	0.1779***	0.1779***
	(0.0070)	(0.0013)
tract_rolling_mean_price_2yr	0.0000***	0.0000***
	(0.0000)	(0.0000)
Age	-0.0046***	-0.0046***
	(0.0003)	(0.0000)
HMSTD_Property	0.0495***	0.0495***
	(0.0037)	(0.0006)
log_lnd_sqft	0.0829***	0.0829***
	(0.0070)	(0.0006)
tract_median_size	-0.0002***	-0.0002***
	(0.0000)	(0.0000)
sale_quarter_cos	0.0050***	0.0050***
	(0.0005)	(0.0004)
IMP_QUAL	0.0233***	0.0233***
	(0.0028)	(0.0004)
pct_college4plus	0.0061***	0.0061***
	(0.0011)	(0.0001)
tract_price_trend	0.2971***	0.2971***
	(0.0538)	(0.0067)
tract_median_age	0.0035***	0.0035***
	(0.0007)	(0.0000)
tract_price_momentum_2yr	-0.0000***	-0.0000***
	(0.0000)	(0.0000)
log_NO_BULDNG	-0.0719*	-0.0719***
	(0.0343)	(0.0083)
Observations	757,278	757,278
R^2	0.8611	0.8611
RMSE (log)	0.2493	0.2493
MAPE (%)	18.8	18.8

Notes: Dependent variable: log(SALE_PRC). Left column: tract-clustered (151 clusters) standard errors. Right column: HC1 robust standard errors.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Table shows top 20 features by absolute t-statistic (tract-clustered specification).

Table 6: Example Valuation Explanation (SHAP Decomposition)

Metric	Value	Notes
Predicted value	$189,528	Point prediction (dollars)
95% Conformal CI	$156,929–$222,127	Out-of-fold interval
Local explanation (selected features)		
Feature	SHAP (log pts)	Approx. impact ($)
log(Living Area)	0.0379	$7,329
LogPopulation	0.0247	$4,745
`recovery`	-0.0216	$-4,049
`years_since_2000`	0.0150	$2,856
`pct_college4plus`	0.0139	$2,652
`sale_quarter_sin`	-0.0128	$-2,407
`tract_price_momentum_2yr`	0.0125	$2,384
`LND_VAL_lag1`	0.0122	$2,335
Age	0.0119	$2,266
`tract_median_LND_VAL`	-0.0109	$-2,062
`tract_rolling_mean_price_2yr`	-0.0089	$-1,681
`IMP_QUAL`	0.0085	$1,622
`TV_SD_lag1`	-0.0071	$-1,344
`AV_SD_lag1`	-0.0064	$-1,204
log(Land Sq Ft)	-0.0060	$-1,125

Notes: SHAP values are additive contributions on the model output scale (log dollars). "Approx. impact" converts each SHAP value to dollars using a local linearization around the point prediction; interpret as a rough magnitude. Background set is balanced by tract and time period.

Table 7: Model Performance Comparison: CQR vs Hedonic OLS

Metric	CQR	Hedonic OLS
R^2 (log scale)	0.968	0.861
RMSE (log scale)	0.120	0.249
MAPE (%)	8.8	18.8
Coverage (%)	93.0	–
Relative Width (%)	34.4	–

Notes: Performance metrics from out-of-fold predictions. CQR uses StratifiedKFold by price quintiles (5 folds). Hedonic OLS is a CQR-comparable log-linear benchmark; inference is reported with tract-clustered and HC1 robust standard errors (see Table 5).

Table 8: Assessment Quality Standards (IAAO)

Metric	Value	IAAO Standard/Target
Median Assessment Ratio[a]	0.000	0.90–1.10
Coefficient of Dispersion (%)[c]	58.9	$\leq$ 5% (newer), $\leq$ 10% (older), $\leq$ 20–25% (rural)
Price-Related Differential	1.516	1.03–1.03
Price-Related Bias[b]	—	Slope ≈ 0
Supplementary Metrics (not IAAO standards)		
Within ±10%	0.0%	Internal Benchmark
Within ±20%	0.0%	Internal Benchmark
Within ±25%	0.0%	Internal Benchmark

[a] IAAO prefers the **median ratio** over the mean as the primary measure of level.
[b] Price-Related Bias is a newer alternative to Price-Related Differential, introduced in IAAO's 2013 *Standard on Ratio Studies.* [c] Coefficient of Dispersion standards vary by property type: newer homogeneous residential properties have stricter thresholds than older/mixed areas or rural/income properties. Notes: Coefficient of Dispersion measures assessment uniformity (lower is better). Price-Related Differential and Price-Related Bias measure regressivity/progressivity; acceptable ranges minimize systematic bias. Supplementary compliance rates (e.g., ±10%, ±20%) are **not official IAAO standards** but may be used by jurisdictions as internal benchmarks.

Table 9: Conformal Coverage Analysis

Metric	Value
Expected Coverage (90% CI)	0.950
Actual Coverage (OOF)	0.975
Coverage Error	0.025
Half-Width (Log Points)	12.9117
Duan Correction Factor	125241.4931
Mean Interval Width ($)	96,568,426,496
Median Interval Width ($)	97,275,576,320

Notes: Conformal coverage using out-of-fold residuals. 90% confidence intervals (alpha = 0.05). Symmetric intervals $\hat{y}$_log $\pm$ q.

Table 10: Conformal Coverage by Predicted Value Quintile

Quintile	N	Coverage	Mean Width ($)	Median Width ($)
Q1	200	0.935	78,559,567,872	81,382,621,184
Q2	200	0.960	91,186,241,536	91,260,108,800
Q3	200	0.985	97,425,588,224	97,275,576,320
Q4	200	0.995	103,418,847,232	103,414,415,360
Q5	200	1.000	112,251,887,616	110,859,419,648

Notes: Conformal coverage using out-of-fold residuals. 90% confidence intervals (alpha = 0.05).

Table 11: Fairness Analysis by homestead

Group	N	MAPE	MdAPE	Mean Log Error
0	476	1.000	1.000	11.5158
1	524	1.000	1.000	11.5005

Notes: Fairness analysis by property characteristics. Errors computed from out-of-fold predictions.

Table 12: Fairness Analysis by Predicted Value Quintile

Quintile	N	Coverage (%)	Mean Width ($)	MAE ($)	MAPE (%)	Bias (%)
Q1 (Lowest)	151,456	91.0%	$35,259	$9,660	14.2%	1.2%
Q2	151,455	93.4%	$39,314	$10,417	7.7%	0.3%
Q3	151,456	94.3%	$52,757	$13,309	6.8%	-0.1%
Q4	151,455	94.2%	$69,186	$17,379	6.4%	-0.5%
Q5 (Highest)	151,456	92.2%	$185,856	$46,972	8.8%	0.2%

Notes: Error metrics by predicted value quintile computed from out-of-fold CQR predictions. MAPE indicates mean absolute percentage error; Bias measures mean signed percentage error. Coverage well-balanced across quintiles (91–94%), indicating fair treatment across predicted value range.

Table 13: Model Performance Comparison: FULL vs. PURE_AVM Specifications

Metric	**Model 1 (FULL)**	**Model 2 (PURE_AVM)**	**Difference**
R^2 (log-space)	0.9679	0.9560	−0.0119
R^2 (dollar-space)	0.9585	0.9460	−0.0125
MAPE (%)	8.81%	10.6%	+1.79 pp
RMSE ($, thousands)	36.0	43.2	+$7.2
MAE ($, thousands)	19.5	24.8	+$5.3
Coverage (CQR, %)	93.0%	92.8%	−0.2 pp
Relative Width (%)	34.4%	40.4%	+6.0 pp
Mean Width ($, thousands)	76.5	89.7	+$13.2
Feature Count	34	26	−8

Notes: All performance metrics from out-of-fold stratified cross-validation (5 folds by price quintiles). PURE_AVM excludes all lagged assessor value components (JV_lag1, LND_VAL_lag1, AV_SD_lag1, TV_SD_lag1, JV_to_LND_VAL_ratio, tract_median_JV, tract_mean_JV, tract_median_LND_VAL). Width values are relative to predicted sale price.